

METHANE



Dr. D. P. Mishra

- This is commonly known as **firedamp** though, technically, a CH_4 -air mixture is sometimes referred to as firedamp.

There are three primary reasons for giving particular attention to methane.

- First, **it is the naturally occurring gas** that most commonly appears in mined underground openings.
- Secondly, **it has resulted in more explosions and related loss of life** than any other cause throughout the recorded history of mining.
- The third reason for giving special attention to CH_4 concerns the **continued development of methane drainage technology**.

Formation of Methane

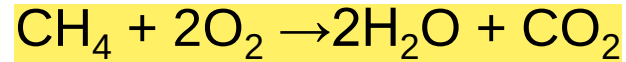
- CH_4 is formed in the **coalification process** of plant materials through biochemical decay and metamorphic transformation.
- CH_4 is a product of decay of cellulose $(\text{C}_6\text{H}_{10}\text{O}_5)_n$ and is formed whenever vegetable matter such as timber decomposes under water and out of contact with air, as in marshes. Hence it is also called **marsh gas**.
- Since it is produced by bacterial and chemical action on organic material, it is evolved during the formation of both coal and petroleum.
- It forms part of coal seams and associated strata as coal has been formed million years ago in that manner.
- The CH_4 content of coal seams increases with depth.
- CH_4 is emitted not only in coal mines but is also found in rock salt, potash and clay mines.

- CH_4 is retained within fractures, voids and pores in the rock either as a compressed gas or adsorbed on mineral (particularly carbon) surfaces.
- When the strata is pierced by boreholes or mined openings, the gas pressure gradient that is created induces migration of the methane towards those openings through natural or mining-induced fracture patterns.
- The gas content of a coal bed is made of two parts –
 - free gas compressed in pore spaces of the coal and
 - the gas absorbed in the internal surface of the coal.
- The proportion of free gas in the coal pores depends on the porosity of the coal seam, gas pressure and temperature. It usually accounts for a small portion 5 to 10% of the total gas content of the seam.
- The greater part of CH_4 is held in situ on the surfaces of the coal pores and microfractures in adsorbed form.
- Since the internal surface area of coal can be as large as $90 \text{ m}^2/\text{g}$, the quantity of adsorbed gas can be extremely high.

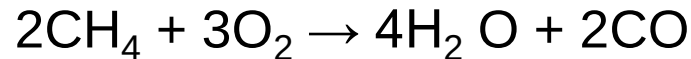
Properties of Methane

- It is a colourless, odourless and tasteless gas.
- Although CH_4 itself has no odour, it is often accompanied by traces of heavier hydrocarbon gases in the paraffin series that do have a characteristic oily smell.
- CH_4 is lighter than air (Sp. gr. 0.554).
- It has a density of 0.68 kg/m^3 at 15°C and is 0.554 times as heavy as air. Due to this it tends to rise to the roof of a mine working and forms pools or layers along the roofs and in rise workings of underground openings.
- CH_4 is not toxic but is particularly dangerous because it is flammable and can form an explosive mixture with air.
- It is combustible but does not support combustion.
- CH_4 burns in air with a pale blue flame.

- In an abundant supply of air, the gas burns to produce water vapour and CO₂.



- Within the confines of mine openings and during fires or explosions, there may be insufficient O₂ to sustain full combustion, leading to formation of the highly poisonous CO.

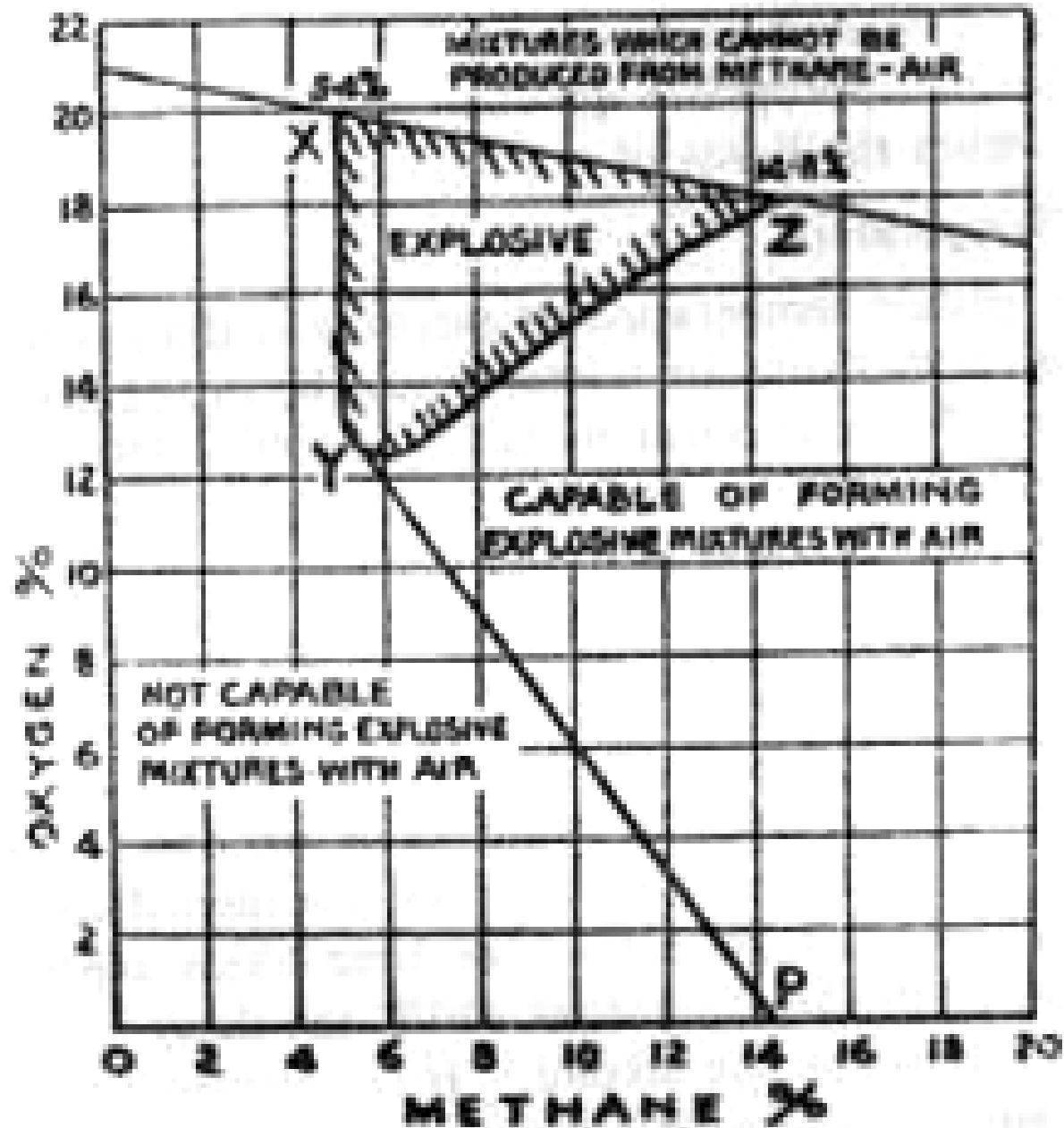


- When mixed with air it forms an explosive mixture, the limits of explosibility being 5 and 15% by volume.
- The gas is not poisonous but suffocates a person due to lack of O₂ if present in large quantities.
- It is hardly soluble in water, 100 vols. of water at 20 °C dissolving only 3.3 vols. of CH₄.

Limits of Flammability, Flammable Limits, or Explosive Limits of Methane

- The flammable limits of CH_4 -air mixtures are the limits of concentration of CH_4 in air between which a flame can be propagated throughout the mixtures.
- The boundary-line mixtures with minimum and maximum concentration of CH_4 in air, which if ignited, will just propagate flame are known as the **lower and upper flammable or explosive limits**.
- - The lowest % of CH_4 in air that yields an inflammable mixture is called the **lower flammable limit (LFL)**.
- - The highest % of CH_4 in air that yields a similar mixture is called the **upper flammable limit (UFL)**.

Coward's Diagram (H. F. Coward, 1928)



It shows the limits of explosibility with different percentages of firedamp and oxygen. The important points to note are:

- All mixtures lying within the triangular area XYZ, are in themselves explosive.
- All mixtures lying to the right of PYZ contain too much methane to explode, but they will form explosive mixtures when mixed with the right amount of air.
- All mixtures lying to the left of PYX are neither explosive, nor capable of forming explosive mixtures with air.
- Lower limit of explosibility remains almost constant at about 5.4% for all percentages of oxygen down to about 12.5%.
- The higher limit of explosibility gradually decreases from 14.8% to about 6% with decreasing percentage of oxygen.
- No percentage of firedamp is explosive when the percentage of oxygen is 12 or less.
- A firedamp-air mixture may become explosive when diluted with an appropriate quantity of air which brings the new mixture within the limits of the triangle XYZ.

Measurement of methane content of coal seam

- The main objectives of determining the gas content of virgin coal beds are:
 - To forecast/predict the expected methane levels in mine air of a prospective mine located in the same coal seams.
 - To assess the viability of methane drainage technique.
- Methane content of coal is defined as the vol. (m^3) of gas in unit mass (t) of virgin coal under natural conditions.
- It depends on
 - The degree of coalification or rank of coal: higher rank coals containing more methane.
 - Depth of the seam and
 - Duration of denudation cycle

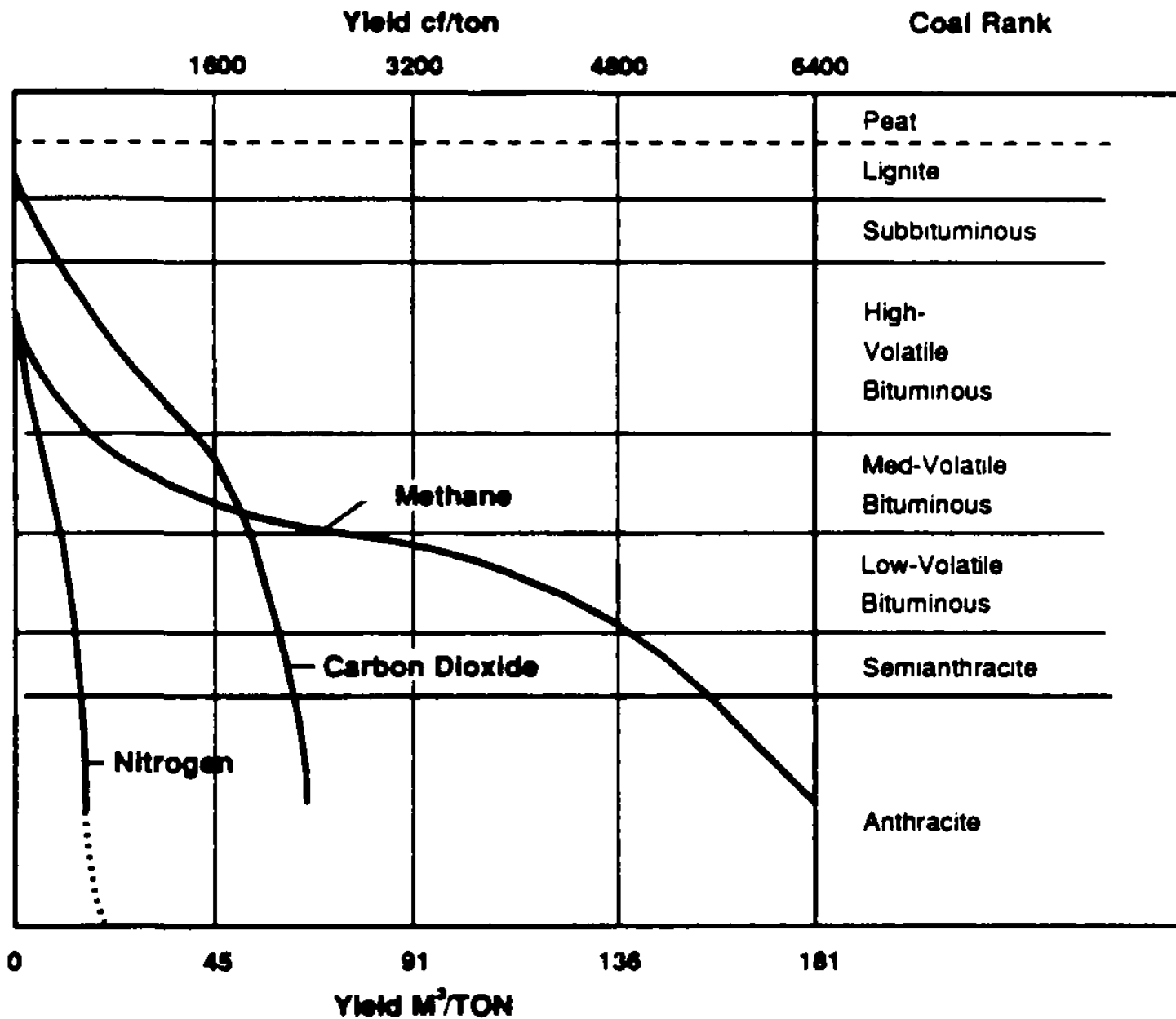


Fig. Gas quantities generated during coalification

- Graham found the gas content of small lumps of coal to range from 0.36 to 16.8 m³/t.
- Russian work shows that gas content of coal ranges upto 30-40 m³/t, occasionally rising to 50-60 m³/t in vary porous and fissured coals.

Methods of estimation of gas content of coal seams

All the methods can be grouped into two categories:

1. Direct methods: The best possible methods.
2. Indirect methods

Direct method of measurement of methane content

- The most commonly used direct method was developed by USBM and hence called **US Bureau of Mines' method**.
- It consists of
 - obtaining a core sample from the seam by drilling and enclosing it as soon as possible in a bomb (sample container) and
 - measuring the gas released with time.
- Since variations in gas content of the coal seam may occur due to variability in the quality of coal, it is preferable to take a full length of the core.

- Sample core is put in the sample container and the lead of the container is sealed and the valve is opened.
- Each container should be capable of holding about 2 kg of core and 35 to 40 cm in length.
- The seals on containers must be capable of holding a gas pressure of 350 kPa without leaking.
- Readings of the volume emitted are started as soon as the coal sample is placed in the container.
- The volume of the gas emitted is continuously recorded at every **15 min** for the next several hours until the gas emitted from the sample is less than **0.05 cm³/g per day for 5 consecutive days**.
- The most convenient way to measure methane emission from the sample container is **water displacement method** in which the water of the inverted graduated cylinder is displaced by the methane gas released from the sample core.

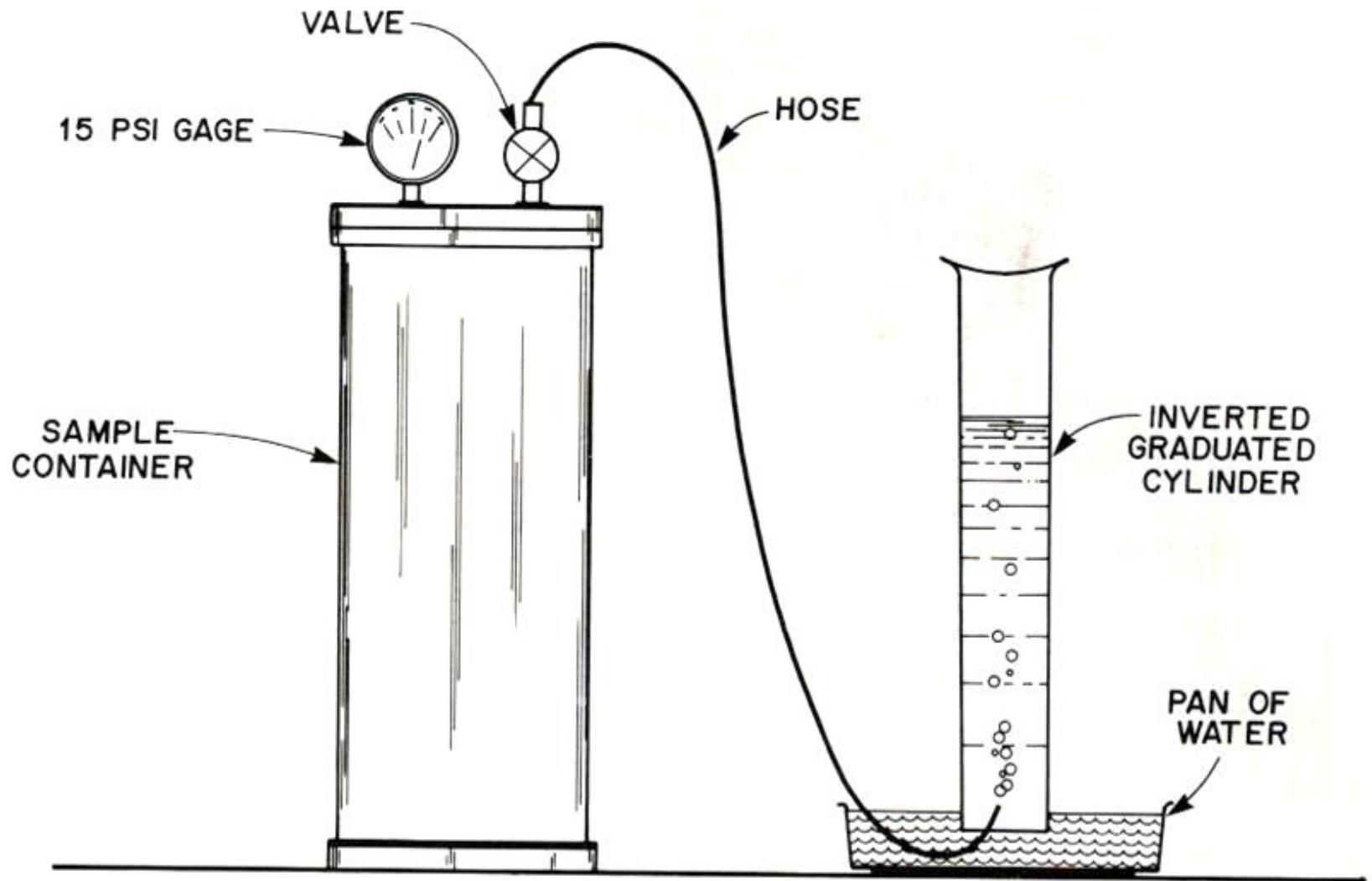
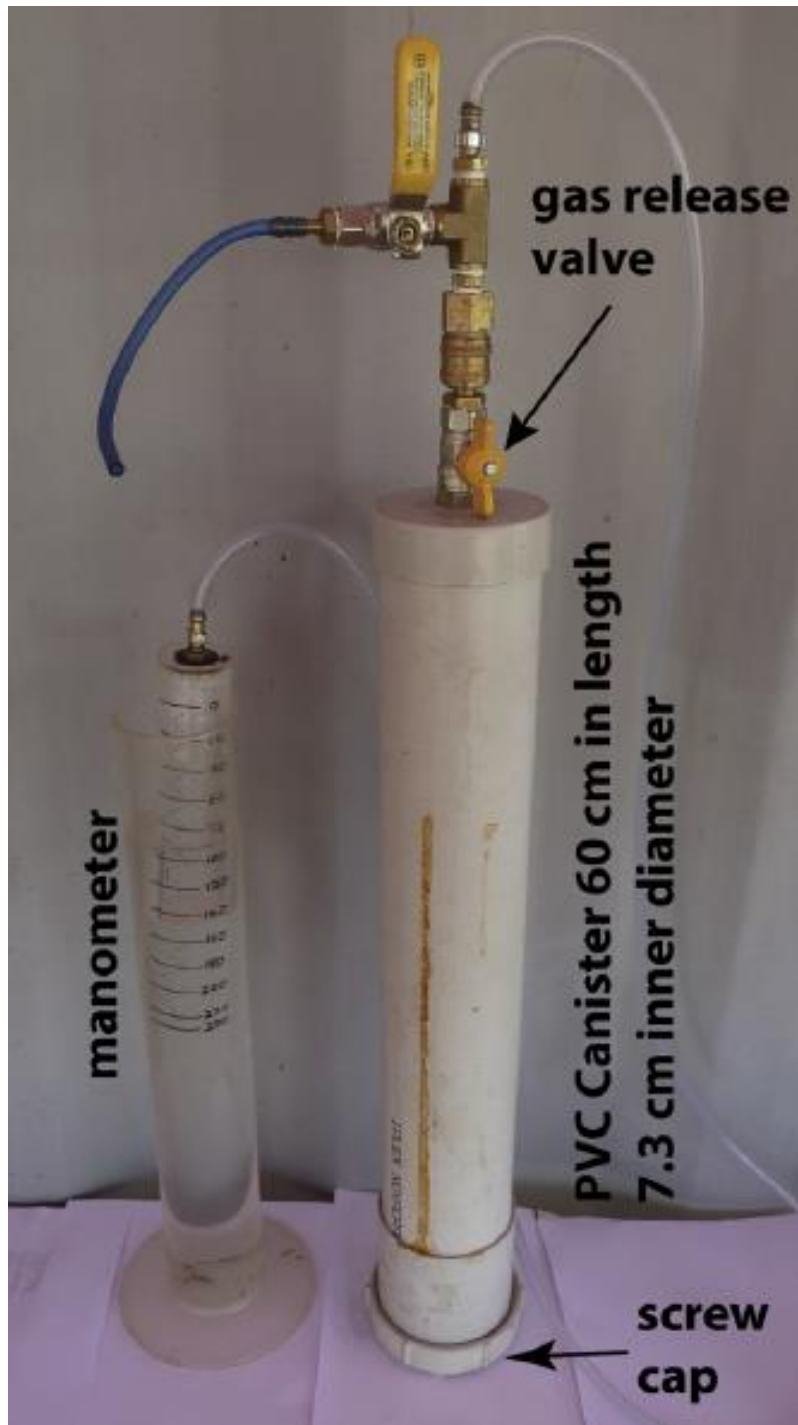


Fig. Sample container
(Gas is measured by displacement of water)



Photograph of typical PVC desorption canister and manometer used in the measurement of desorbed gas from core.

- After desorption is completed or virtually completed, the core is taken out of the bomb and its mass is determined.
- Then the core is crushed to at least 200 mesh in a sealed bomb using steel rods or steel balls and the gas released during crushing is measured.

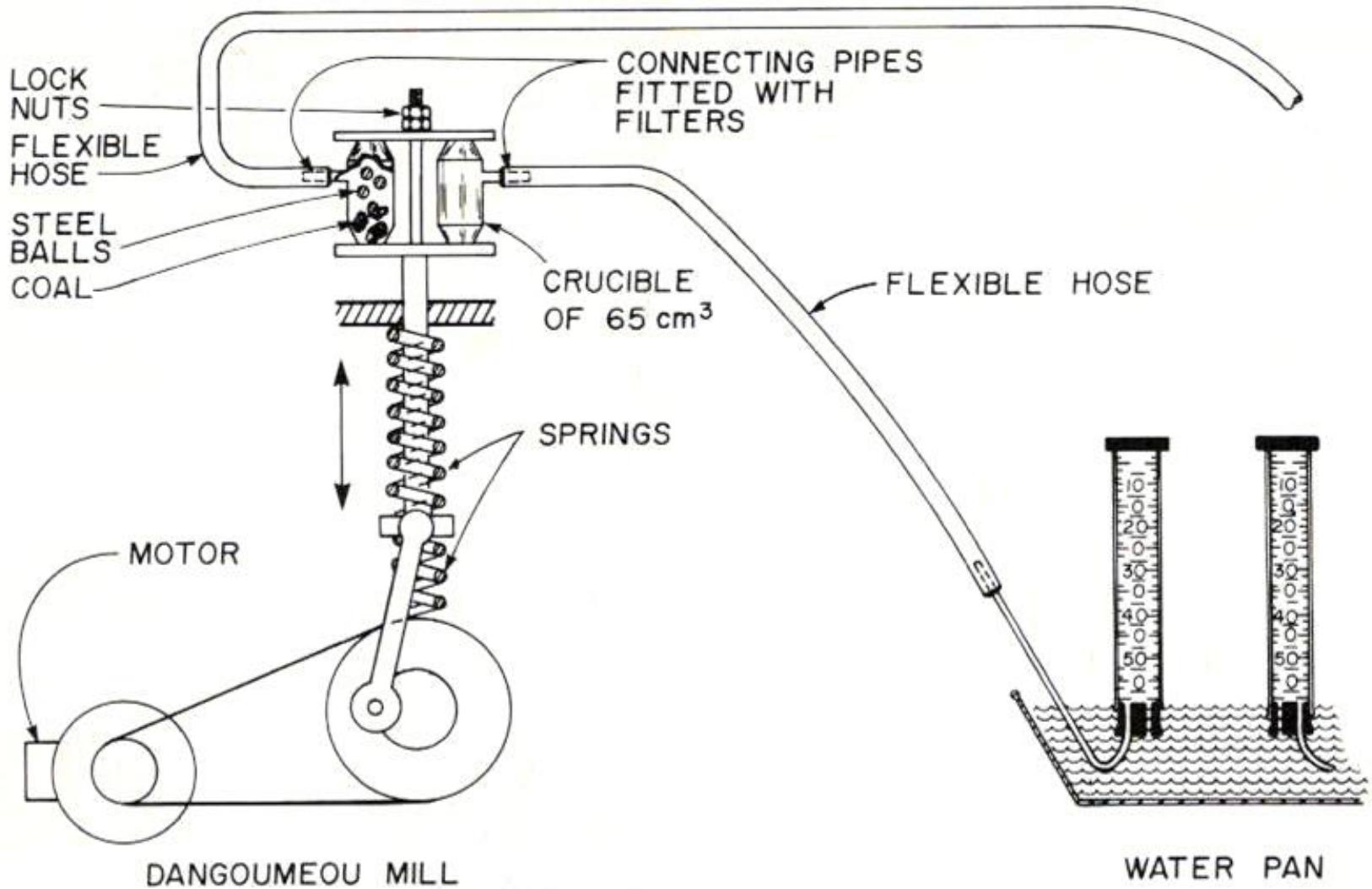


Fig. Gas content determined by crushing coal sample



Photograph of ring mill, manometer used in the pulverisation of coal and measurement of residual gas (Q3).

- The total gas content of the core Q is obtained using the relationship

$$Q = Q1 + Q2 + Q3$$

$Q1$ = gas lost during the time of core is being removed from the borehole and placed in the sample container called **lost gas**.

$Q2$ = gas liberated from the core while in the sample container called **sorbed/desorbed gas**.

$Q3$ = gas liberated when the coal sample is crushed, commonly called the **residual/rest gas**.

Calculation of lost gas

- In estimating the amount of gas lost, lost time has to be established accurately.
- The quantity of lost gas can be determined theoretically by calculation.
- A graphical technique which is more accurate and rapid can also be applied.
- The vol. of gas first given off from the sample container is plotted against $\sqrt{\theta}$ to calculate gas lost during transfer of the sample from seam to bomb.

Where,

$\theta = (t + 150)$ min is the time since CH₄ pressure on the core was released

t = time measured from the instant coal is inserted in the sample container

150 min is the time elapsed between this instant and coring of coal.

By extrapolating the linear portion of the graph back to $\theta = 0$, **the lost gas** is obtained.

Some investigators believe that determination of lost gas is not essential because of the overall inaccuracy of the system and take the lost gas equal to 10% of $(Q_2 + Q_3)$.

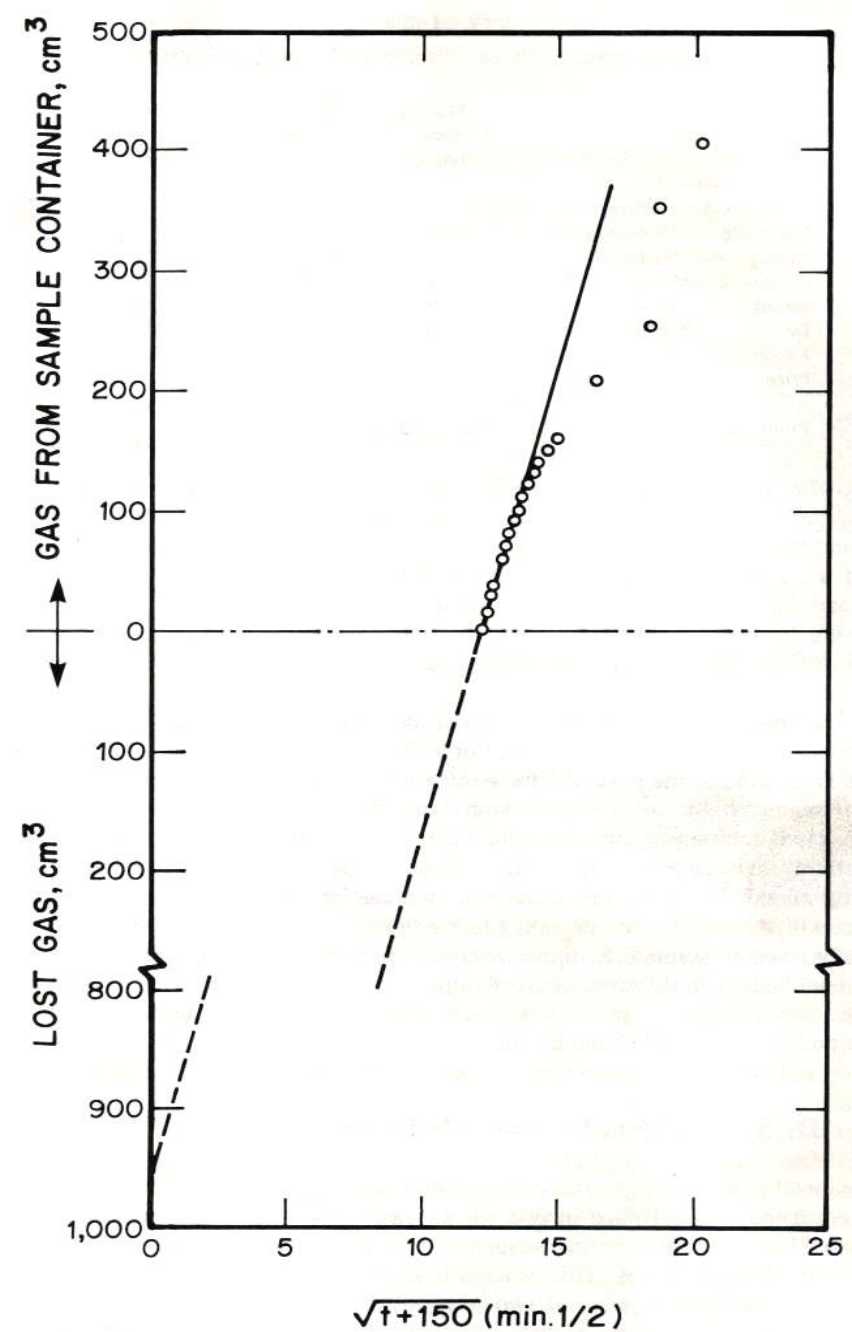
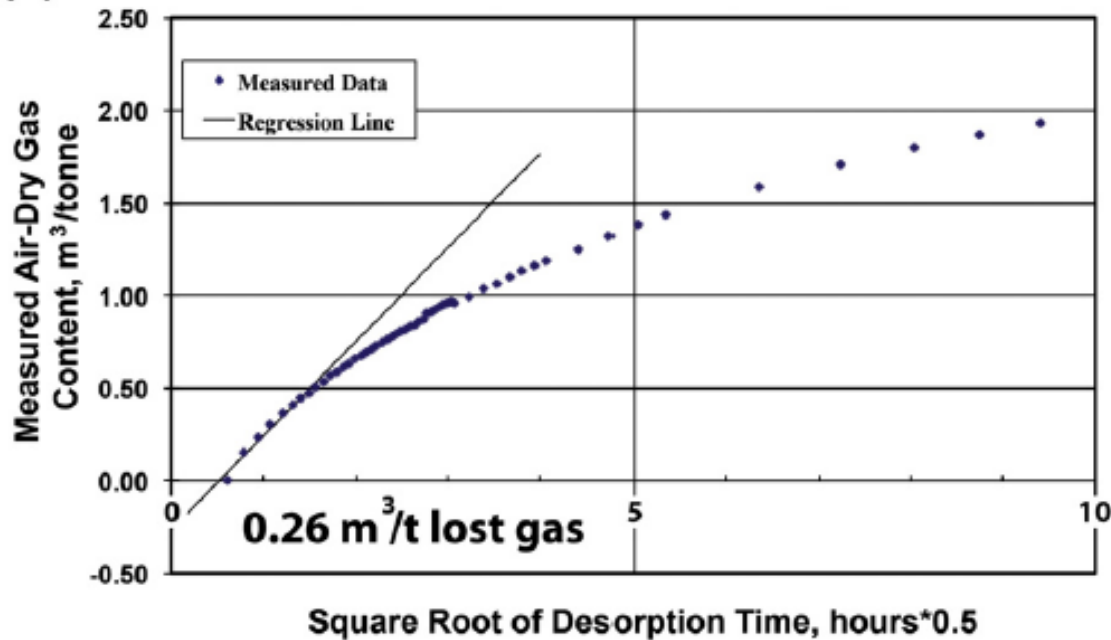
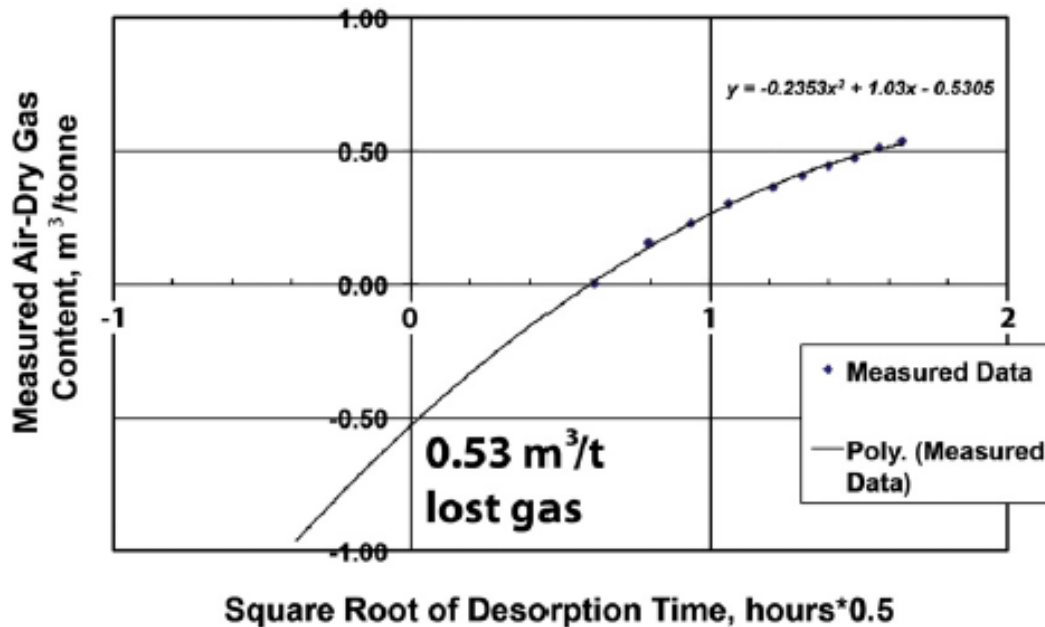


Fig. Emission rate curve

(A)



(B)

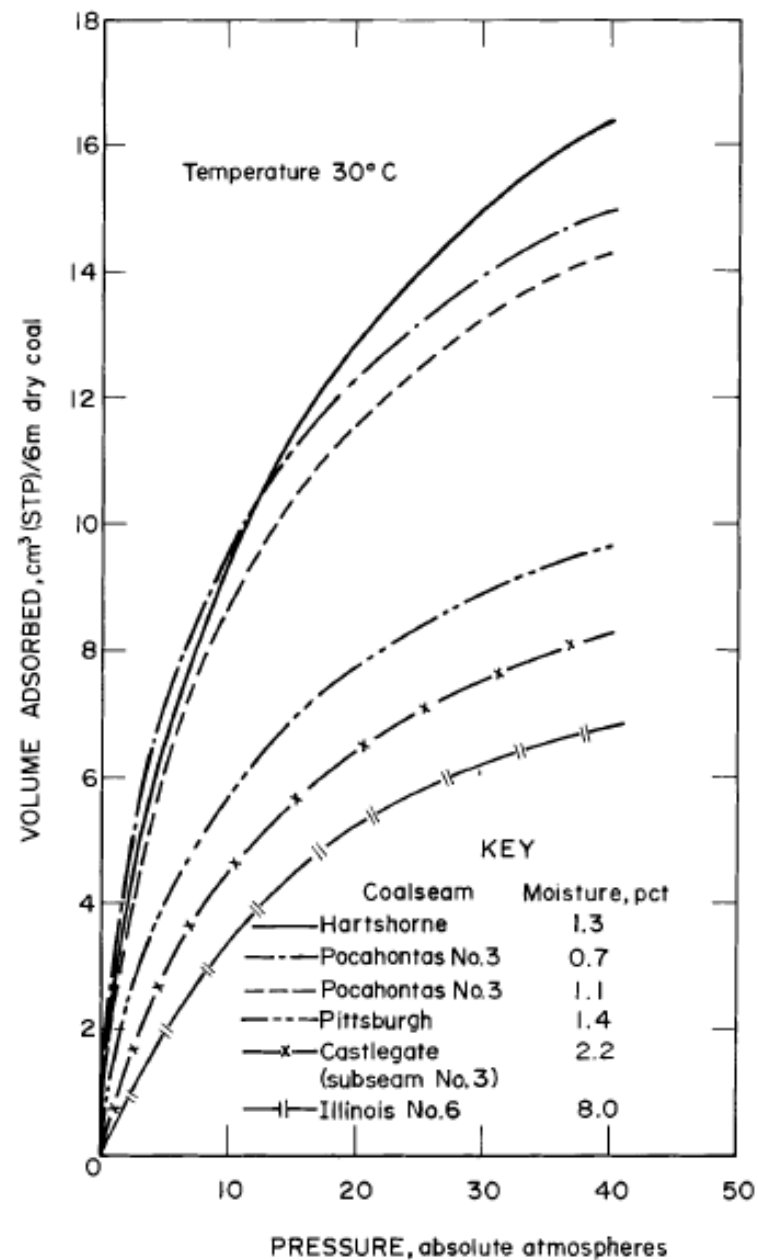


Example of (A) a linear regression indicating $0.26 \text{ m}^3/\text{t}$ of lost gas (Q1) and (B), on the same data, a second order polynomial regression line indicating $0.53 \text{ m}^3/\text{t}$ of lost gas.

Which is the correct way of estimating lost gas is still a debated topic.

Indirect method of measurement of methane content

- An indirect method is sometimes used to estimate the methane content in virgin coal beds.
- In indirect method, as per USBM measurements, a vertical borehole is drilled into the virgin coal seam from the surface.
- The shut-in pressure in boreholes drilled into coal is measured which is correlated with the methane content by means of an empirical curve which is determined experimentally in the laboratory.



Equilibrium adsorption isotherms: The laboratory relation between the amount of gas adsorbed and the pressure

- From the known reservoir pressure, the gas content can also be estimate using Langmuir's equation [Langmuir, 1918]:

$$V = V_m B P / (1 + B P)$$

Where

V = estimated gas content of coal,

V_m = volume of gas for full saturation of coal,

B is a characteristic constant of the coal seam, and

P is the reservoir pressure.

For U.S. coalbeds, the reservoir pressure is roughly correlated with the depth of the coal seam [Thakur and Davis, 1977] and is estimated at 0.303 psi/ft, or roughly 70% of the hydrostatic head.

Emission of methane

In coal mine, methane may find its way in the workings in the following ways:

- **Gradual exudation or bleeding** from the coal and adjacent strata in the roof and floor.
- **In the form of blowers:** Can be felt on the hand and may in some cases be heard.
- **In the form of gas outburst:** May sometimes associated with violence.
- **Release by roof fall or sudden fall of barometric pressure** which may force the gases of the goaf into the workings.

Gradual exudation of methane

- **From exposed coal surfaces:** CH_4 is emitted chiefly by a process of slow exudation from the seam and adjacent strata when fresh surfaces of coal and other CH_4 -bearing formation are exposed during the process of working.
- **From mined coal:** Pulverization of coal during cutting, loading, transportation etc. causes desorption of methane.
- The size of coal produced has a great influence on the amount of gas emitted at the face.
- While large lumps release only 2% of their total gas content in the 1st ten minutes, coals of 0.25-1 mm size release 40% and of < 0.25 mm size, 66%.
- However, the exposed coal surface after mining produces the major part (50-75%) of the gas rather than the mined coal.

Gas blowers

- Gas is emitted from strata in the form of continuous blowers or feeders, sometimes at a fairly high pressure, for few minutes to several years.
- A gas blower is a powerful irruption of gas in the form of jets from cracks or coal faces.
- It is accompanied by hissing or roaring sound which can be heard over a long distance in the calm of an underground mine.
- Blowers are usually met in areas which have been faulted or folded.
- The gas from a blower is generally pure CH_4 .
- Gas emission from a blower is usually heavy in the initial stages and gradually goes down till it is finally exhausted.
- Gas from permanent blowers can be coursed to the surface and utilized for scientific or industrial purposes, for power generation, lighting or heating.

- A well known example of a blower was in a **British coal mine of Garwood**.
- The blower of very long duration was met in a sinking pit and the gas was led to the surface and burnt for 9 years in a flare so high that the flame could be seen some 15 km away.
- Another example of a gas blower of very long duration is that of **Cymmer Colliery in South Wales (UK)** which has been feeding gas, 97.5% methane at a rate of 25 m³/hr for over 70 years.
- In coal seams which are prone to gas blowers underground roadways should be advanced with advanced exploratory boreholes using special drilling machines.
- The effective way of dealing with blowers in a mine is
 - to seal the area off,
 - blocking or closing the crack from which the blower emanates and
 - coursing the gas of blower to the surface by special pipes.
- However, the ventilation of the district has to be improved to keep the percentage of gas in the general body of the return air within permissible limit.

Gas outbursts

- Sometimes, methane may be given out from the strata in violent outbursts which emit large volumes of gas in a short time along with a lot of small coal and fine dust.
- They are accompanied by a blast of gas and fog.
- Outbursts generally occur in the vicinity of areas of geological disturbance owing to sudden release of confined gas in the disturbed zone as the confining pressure is released by a working approaching the area.
- It occurs from adjoining strata where gas has been trapped as no permeable channel existed for the release of the gas.
- High stress combined with high gassiness of a seam induces an instantaneous outburst.
- Coal is projected violently from the face and disintegrates, releasing the gas.

- The face can be flooded with fine broken coal, a large amount of gas and coal dust.
- The projected coal may cause violence.
- Such outbursts have occurred with differing severity in Australia, UK, Belgium, Canada, Japan, Poland, USSR, Turkey and south Africa.
- The outburst that occurred at the **Valleyfield Colliery, UK in 1911** caused the advance of a level by 3m throwing out about 90 tonnes of small coal and killing 3 men.
- In a **Belgian mine** more than 120 persons were killed due to outburst of CH₄ liberating 3,39,800 m³ gas.
- The gas outburst can be prevented by
 - proper mine planning,
 - conducting pre-mining methane drainage and
 - adopting stress relief techniques.

METHANE LAYERING

- Because of low density, methane has a tendency for streaming/stratifying particularly in steeply dipping roadways and forms horizontal layers.
- Evidence of a no. of gas explosions showed that they were caused by the ignition of these layers.
- When methane is emitted at the roof, travels up the dip in a layer near the roof depending on the velocity of air-current and roughness of the roof.

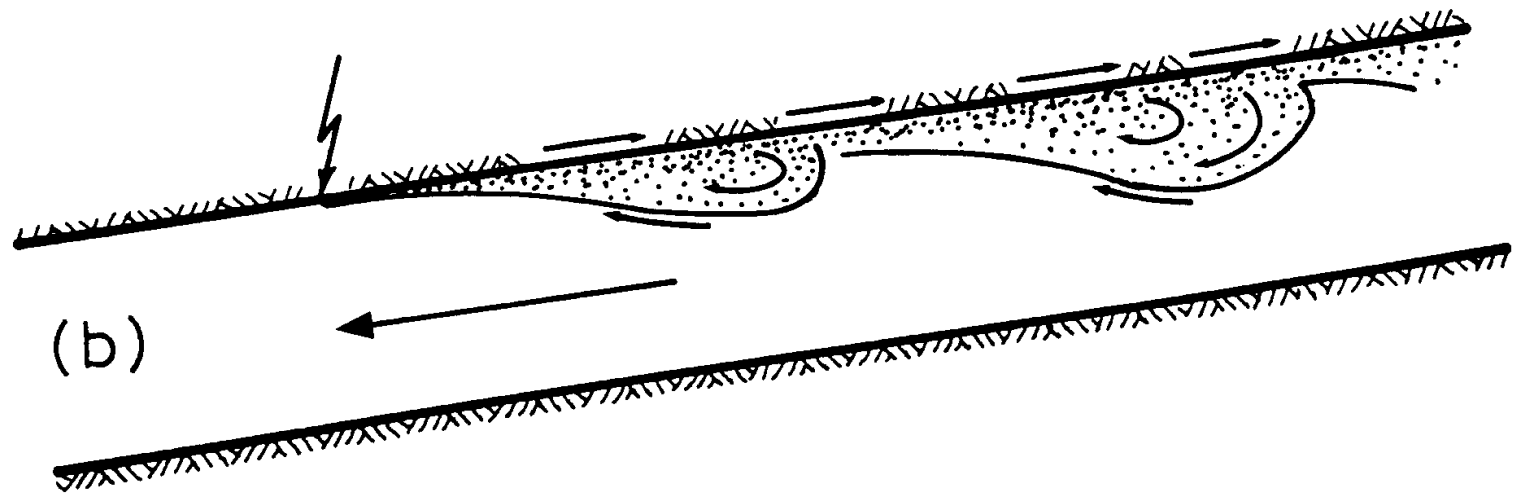
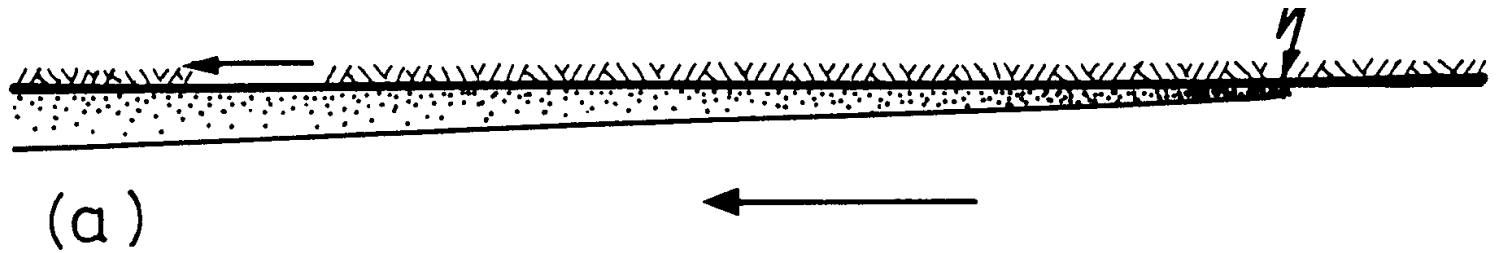


Fig. Methane layering in (a) a level airway

(b) A descentionally ventilated airway

Factors affecting methane layering

Nature of airflow and presence of obstructions in the airways:

- With turbulent airflow and a rough roof with obstructions such as roof bars etc. the tendency of streaming is reduced.

Direction of movement and velocity of air-current:

- If the air-current travels down the dip, it breaks up streaming and carries the methane down the dip.
- The tendency for streaming is greater with an up-the-dip air velocity where the critical velocity becomes higher.
- **The critical velocity** for breaking up of streaming in smooth roadways depends on the dip and varies from 0.04 m/s for a slope of 1 in 100 to 1.45 m/s for a slope of 1 in 2.8.

Stable CH₄ layers can also develop at the roof of horizontal airways depending chiefly on

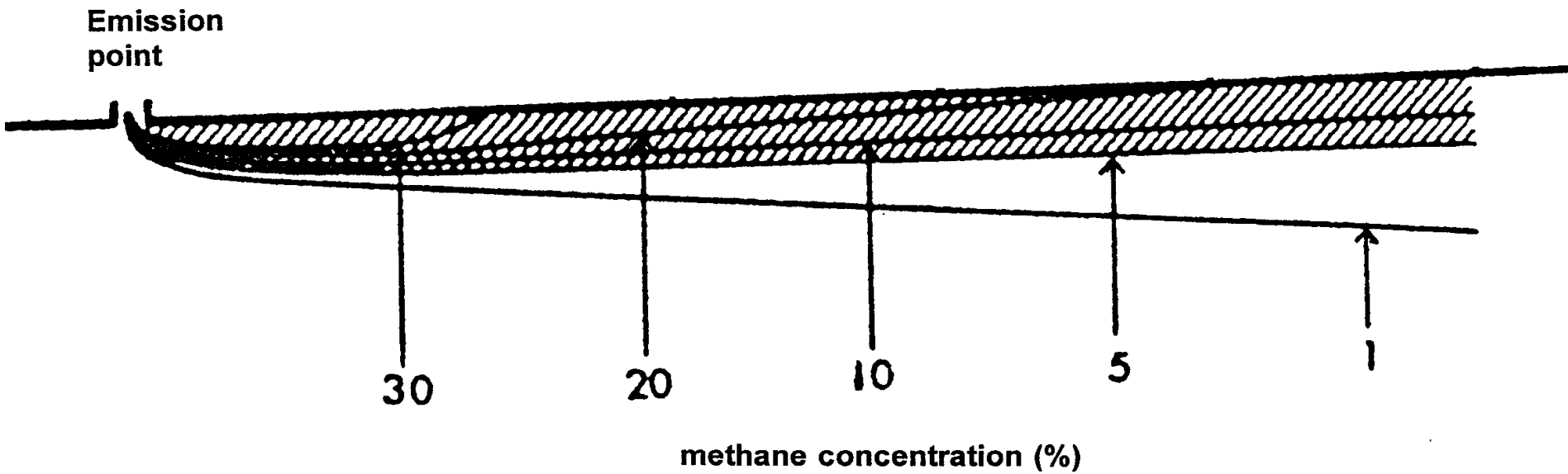
- Rate of gas emission, particularly at the roof
- Size of the airway,
- Velocity of air and
 - Layering of CH₄ occurs when ventilation is insufficient to readily disperse the gas into the air current.
 - In large airways with relatively low average air velocity, there occurs at the roof a laminar boundary sublayer of sufficient thickness.
 - Any CH₄ emitted into this boundary sublayer cannot get diluted by turbulent diffusion and hence forms into a layer sometimes up to 1 m thickness which moves in the direction of the air-current at a slower velocity.

- Other factors affecting methane layering are
 - roughness of the airway surface,
 - presence of bends, obstructions etc. nearby and
 - nature and location of the sources of CH₄ emission.

- The conc. of CH₄ in the layer gradually decreases from the top downwards with almost pure CH₄ occurring against the roof.
- In one case it was nearly 38% gas accumulated at the roof level but its conc. was only 1% at 0.3 m below the roof.

The edge of a layer: defined as the point where the conc. is a certain fraction of the conc. at the roof.

The thickness: defined as the perpendicular distance between the roof and the edge of the layer (Bakke and Leach, 1962).



Methane layering in the roof of a gently inclines roadway

The Layering Number (L)

- **In general, layering of methane depends on**
 - velocity of the ventilating air stream, (m/s)
 - rate of gas emission, (m³/s)
 - width of airway (m)
 - inclination of airway
 - relative densities of the air and gas
 - surface roughness of the roof above the layer (smooth or rough)
 - type of ventilation (ascensional or descensional),
- **A methane layer can be described by combining all these factors into one number, called the layering number (L)**

- The stability of methane layers as well as their length is indicated by the dimensionless layering number.
- **P. Bakke and S. J. Leach (1962)** found that the characteristic behaviour of a gas layer was proportional to the dimensionless group.

$$L = \frac{v}{\left\{ g \frac{\Delta\rho}{\rho} \frac{Q}{W} \right\}^{\frac{1}{3}}} = \frac{v}{(4.37Q/W)^{\frac{1}{3}}}$$

Where

$\Delta\rho$ = difference in relative densities (specific gravity) of air and methane,
i.e. $1 - 0.554 = 0.446$

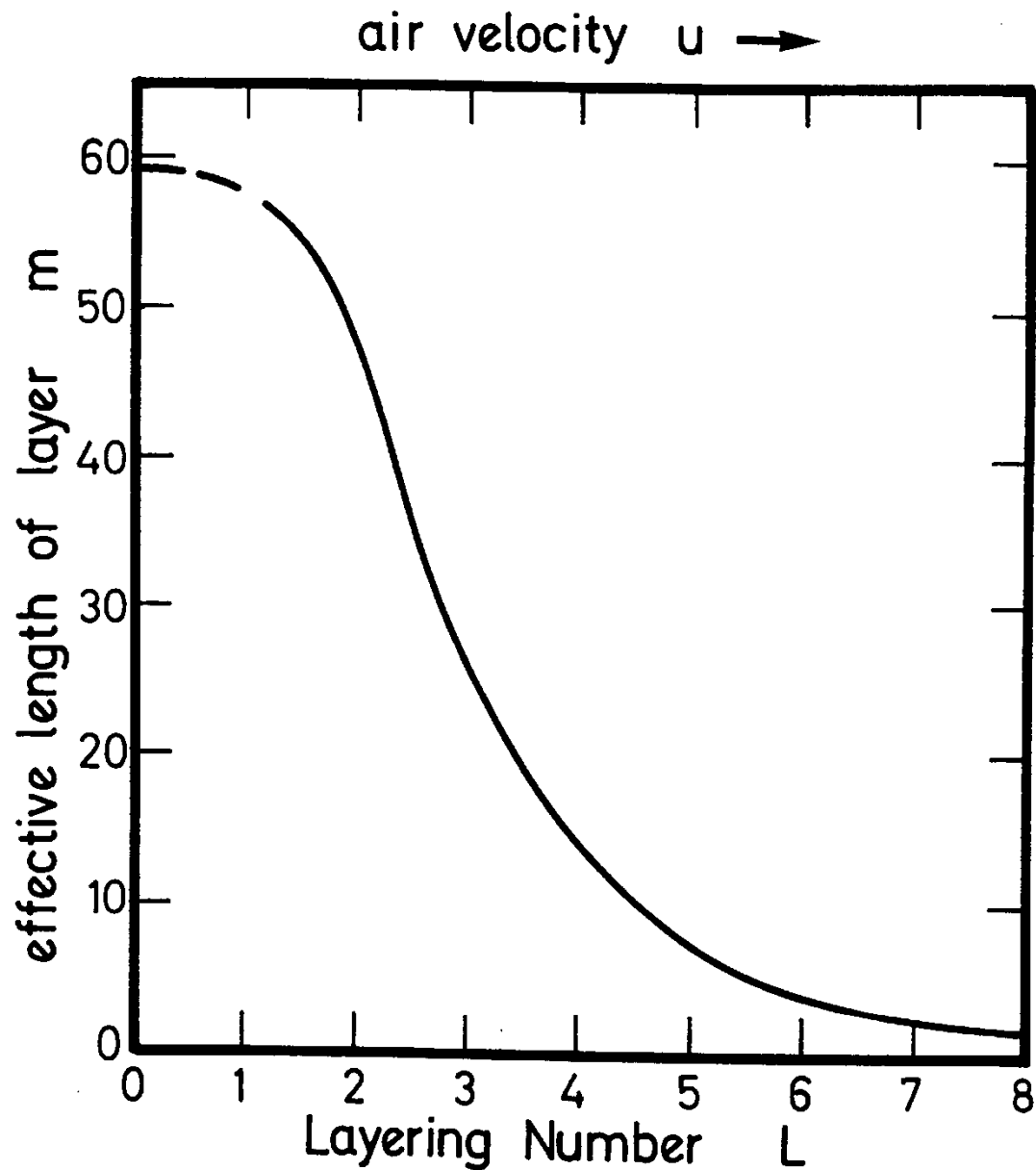
ρ = density of air

v = average velocity of air, m/s

Q = quantity of methane emitted into the roadway, m³/s

W = width of airway, m

The constant 4.37 is the product of g and the difference in relative densities of air and methane to the relative density of air.



It is obvious that CH_4 layering depends much more strongly on the air velocity than on make of gas.

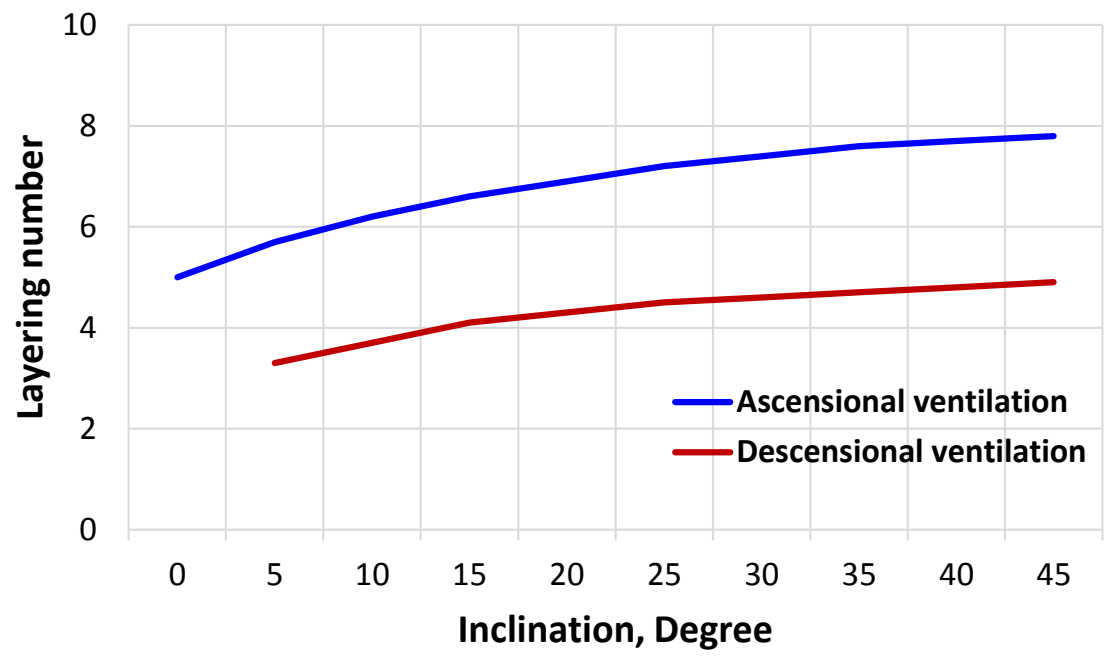
Length of methane layer is the distance from the source to the point where the mean conc. of CH_4 in the layer is 5%.

Fig. Variation of layer length with layering number for level airways

- In a horizontal airway, layering no. exceeding 2 causes turbulent diffusion and breaks up the CH₄ layer. However a layer of sufficient length still forms.
- Generally, there is a sharp decrease in the length of layer with increase in the layering number > 5 and hence should be taken as a desirable value.
- Based on the results, it may be recommended that in level airways, Layering No. should not be less than 5.
- In inclined airways, desirable layering number depends on the direction of airflow.
- For ascensional ventilation, the desirable layering no. is around 8 and for descensional ventilation, it is less than that for horizontal airways.

- In inclined airways, desirable layering number depends on the inclination of airway and direction of airflow.
- For ascensionally ventilated airways, the minimum layering number required is 8 at an inclination of 45° and for descensionally ventilated airways, it is less than that for horizontal airways.

Type of ventilation	Angle to horizontal (degree)									
	0	5	10	15	20	25	30	35	40	45
Ascensional	5	5.7	6.2	6.6	6.9	7.2	7.4	7.6	7.7	7.8
Descensional		3.3	3.7	4.1	4.3	4.5	4.6	4.7	4.8	4.9



Recommended Layering Numbers for preventing methane layering in ascensionally and descensionally ventilated airways of various inclinations

Prevention and remedial measures / Dealing with methane layering

- Problem of methane layering becomes important in highly gassy coal mines particularly where large sized and duplicate airways cause reduction in air velocity.
- Such layers pose a constant explosion hazard and have to suitably dealt with.

The methods to deal with layering are as follows:

- One practical and long-term solution to deal with layering is to increase the velocity or airflow of ventilating air current.

In addition to that other remedial measures are as follows:

- **Placing vertical or inclined hurdle sheets across the airway a little below the roof:** extending hurdle cloth some three quarters of the height of the airway helps in diverting the main air-current to the roof and breaks up the layer.

- **Using venturi blowers:** for breaking up methane layers, where compressed air supply is available.
 - Small 100 mm dia. venturi blowers fed by 3 mm jets of compressed air is sufficiently powerful to break up layers as much as 70m length.
- **By conducting methane drainage:** to reduce gas emission from the strata and minimize the occurrence of CH₄ layering.
- **Controlling / reducing rate of gas emission:** by a suitable selection of mining method, length of face, rate of face advance, type of mechanization, method of ground control etc.
- **Prior working of an adjacent less-gassy seam:** helps in reducing the rate of gas emission.
- **Infusion of the seam ahead of the face with water or foam:** has been used for sealing the pores in coal and preventing gas migration to the face.

METHANE DRAINAGE

- Methane is a rich source of fuel.
- 1 kg of CH₄ in burning evolves 13,600 kcal of heat, whereas, 1 kg gunpowder releases only 580 kcal and 1kg of nitroglycerine gives out 1500 kcal.
- CH₄ has successfully drained from coal seams in many countries and utilized for industrial as well as household purposes.
- The surest way of bringing down the rate of CH₄ emission in highly gassy seams is by methane drainage.
- Apart from minimizing the hazards, it often yields valuable quantities of CH₄ which can be used as fuel.

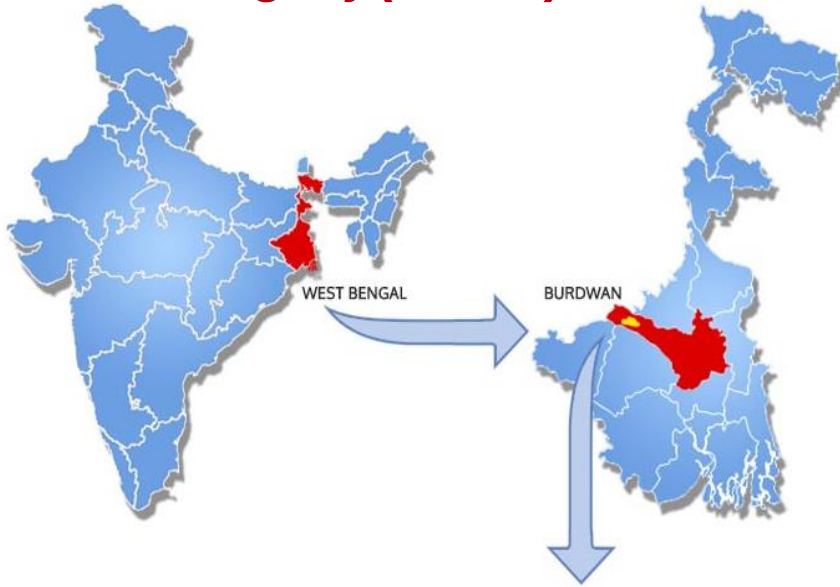
Broad objectives of methane drainage

- To reduce CH₄ emission : **Environmental protection**
- Minimize explosion hazard in highly gassy mines: **Mines safety**
- To produce CH₄, a rich source of energy, for utilization as fuel: **Alternate source of energy**

- It is considered wise to adopt methane drainage in all seams with a gas content exceeding 20-25 m³/t.
- Methane drainage was 1st tried in a colliery of the Ruhr Coalfields, Germany in 1943 and since then it has been successfully adopted in foreign countries.
- In India it is experimented in highly gassy Amlabad Colliery, BCCL during 1975-78 and some more gassy seams are being evaluated for the purpose.
- The reason for not trying in a large scale in India is that our mines are not very deep and not very gassy.

Great Eastern Energy Corporation Ltd.

Raniganj (South) Block



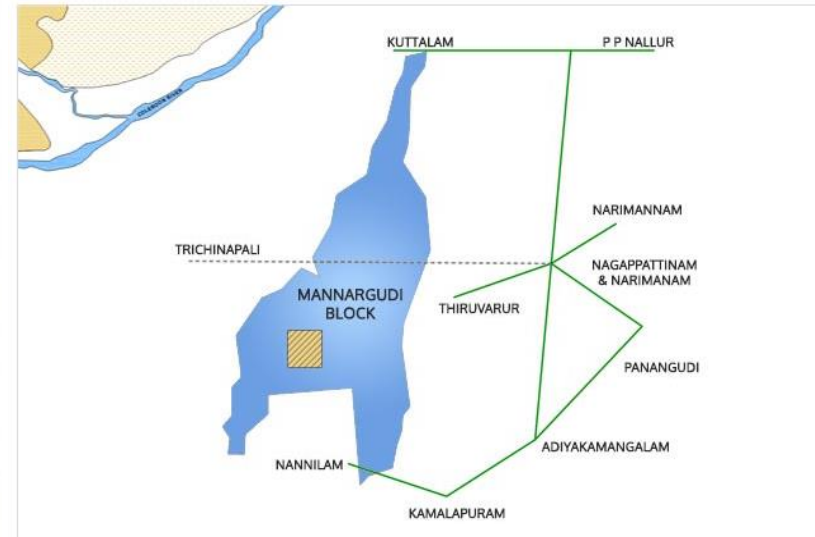
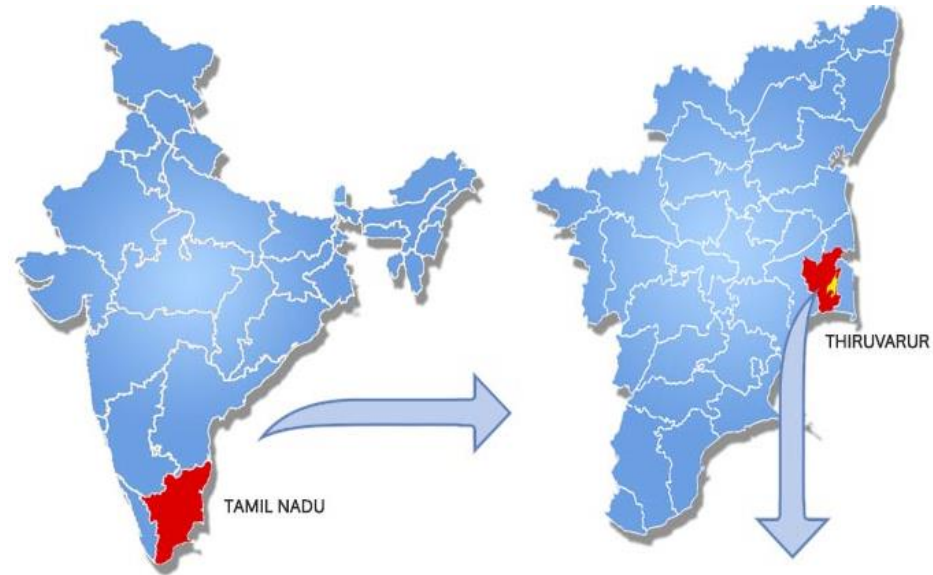
 CBM BLOCK

 GAS GATHERING STATION

 GEECL'S DEDICATED PIPELINE

 CENTRAL GATHERING STATION

Mannargudi Block



 CBM BLOCK

 EXISTING GAS PIPELINE

 PROPOSED PIPELINE BY IOCL

- Degasification during methane drainage can be conducted from
 1. The seam being worked, or
 2. The neighboring seams above or below the seam being worked, depending upon accessibility, or
 3. The goaf.
- In methane drainage, boreholes are drilled from the intake or return airway into the strata which are presumed to be a natural reservoir of gas.
- The boreholes are 45 to 60 degree off vertical leaning towards the in by, 37 to 75 mm in dia and 20 to 60 m apart.
- Hole lengths of 80m are not uncommon. Long boreholes yield generally gas with higher CH₄ content.
- All the holes are equipped with water separators and devices for measuring the pressure and quantity of gas flowing.
- The gas is removed from the boreholes by suction using rotary vacuum pumps installed on the surface.

METHANE DRAINAGE TECHNIQUES

- There is no single preferred technique of methane drainage.
- The major parameters that influence the choice of method include
 - the natural or induced permeability of the source seam(s) and associated strata
 - the reason/purpose for draining the gas
 - the method of mining.

Classification of methane drainage techniques

The common methods being practiced for CH₄ drainage are

1. **Pre-mining methane drainage:** methane drainage performed prior to mining either from virgin seam or from seams being mined.

A. Drainage from extracting seam

- a. **In-seam drainage method:** Gas extraction from seams being mined.
- b. **Cross-measure (vertical or inclined) borehole method**

B. Drainage from adjacent/neighbouring seams

- a. **Superjacent heading method or Hirschbach method:** Gas drainage from adjacent seams

2. **Post-mining methane drainage:**

- a. **Gob drainage by surface boreholes:** Gas drainage from goaf.
- b. **Pack-cavity method.**

In-seam drainage

- Methane flow into mine workings can be reduced significantly by pre-draining the seam to be worked.
- In-seam drainage is successful only if permeability of coal is high (≥ 5 mD). 1 Darcy permits a flow of $1 \text{ cm}^3/\text{s}$ of a fluid with viscosity 1 cP.
- Horizontal holes are drilled into the coal seam from development entries in the mine, which drain CH_4 from the unmined areas of the coal seam shortly before mining, reducing the flow of CH_4 into the mining section.
- Boreholes of even more than 1000 ft length are drilled within the seam using down-the-hole motors and steering/direction-finding mechanisms.

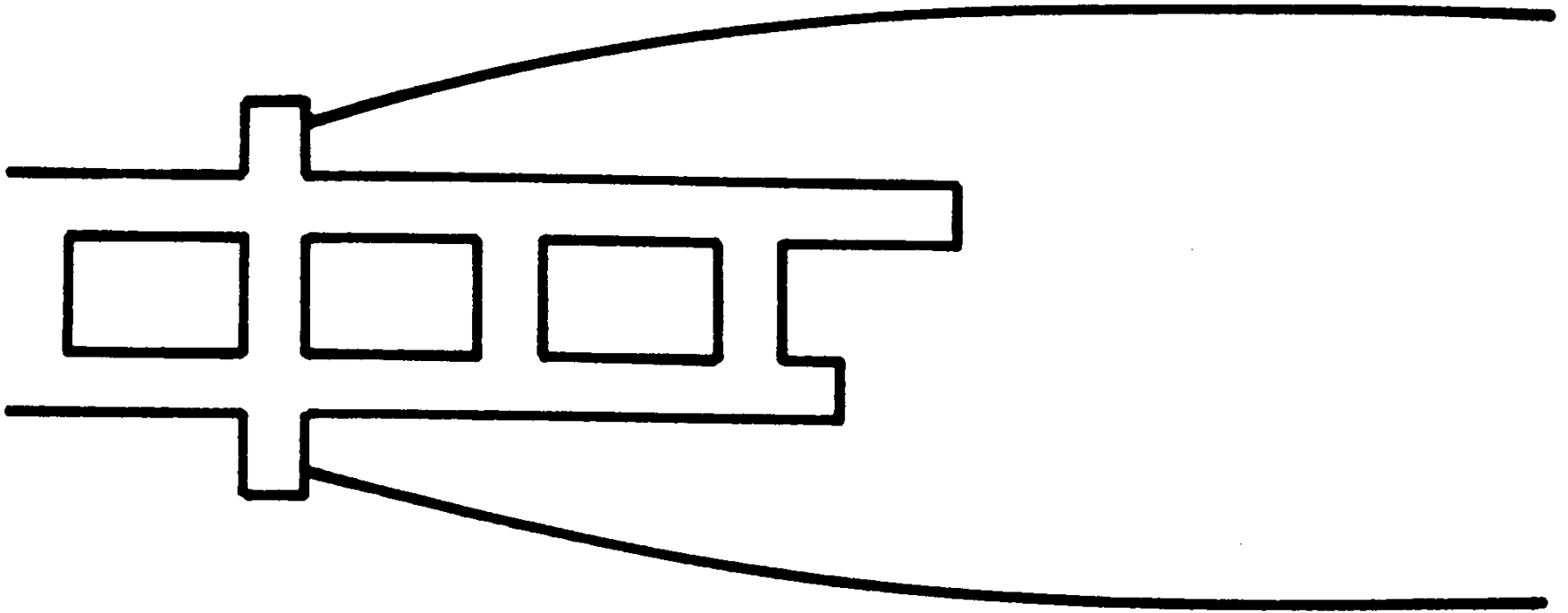


Fig. In-seam drainage boreholes to reduce methane flow into advancing headings

Flanking boreholes used to drain gas from the coal ahead of headings that are advancing into a virgin area.

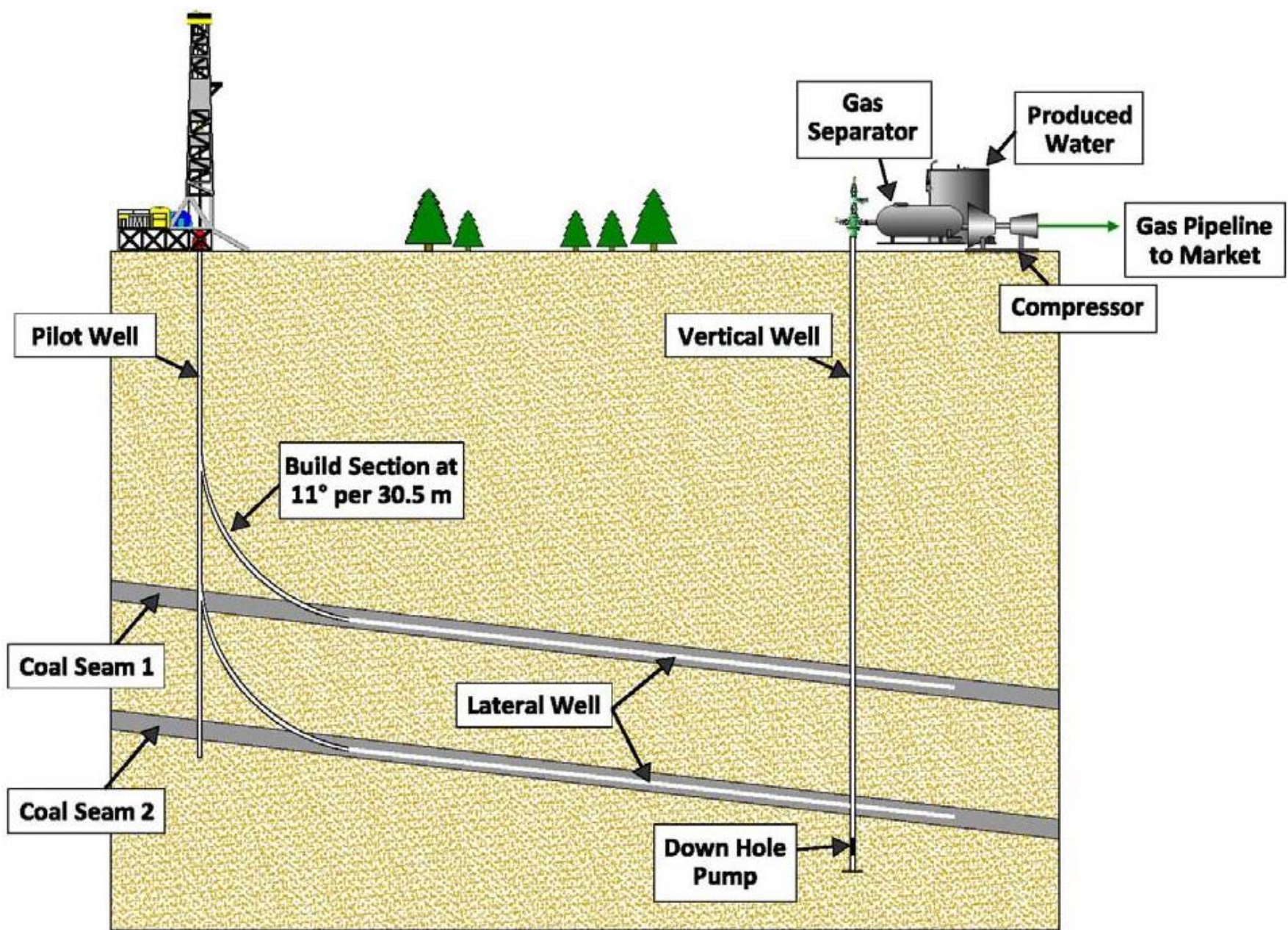
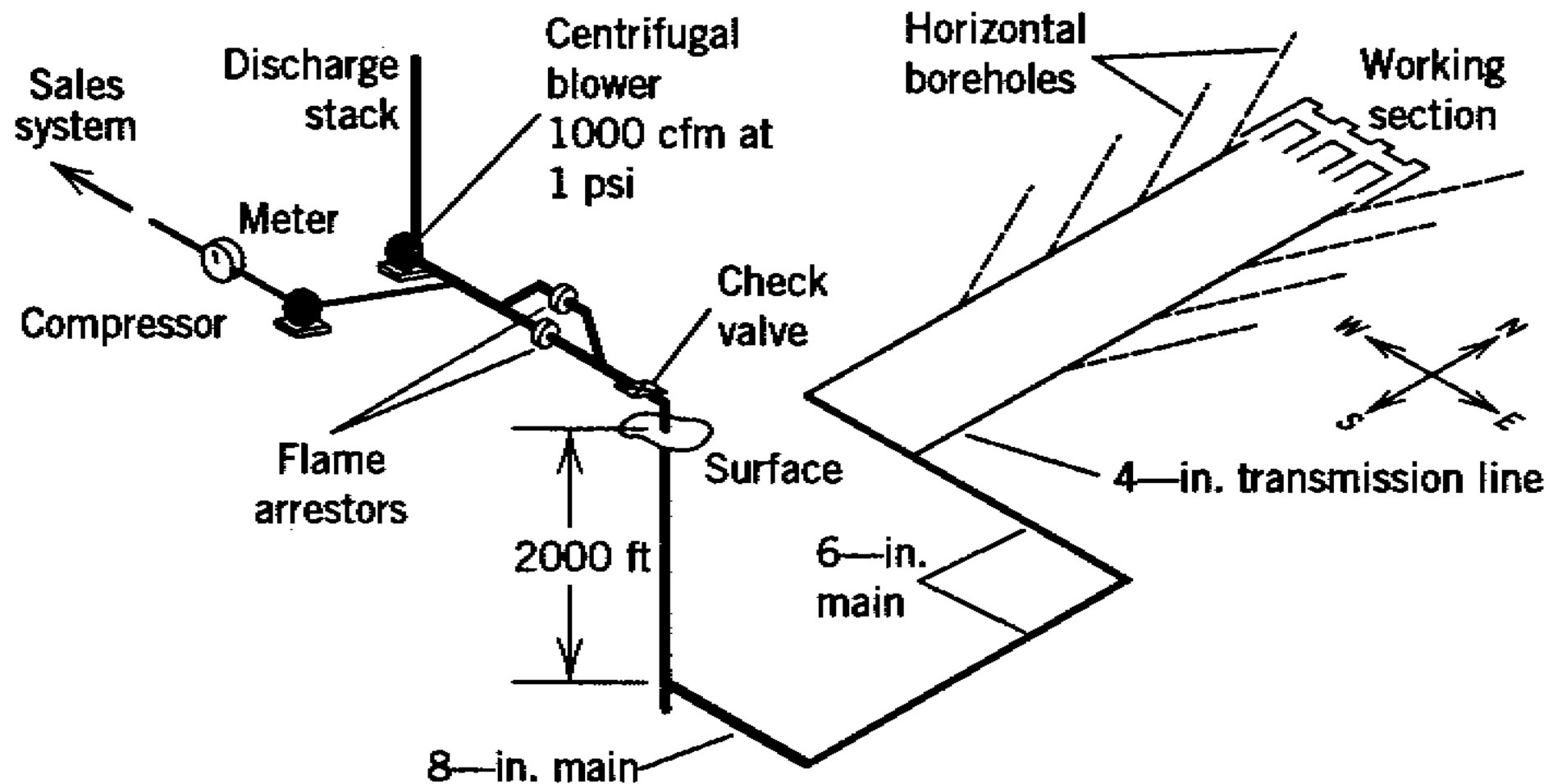


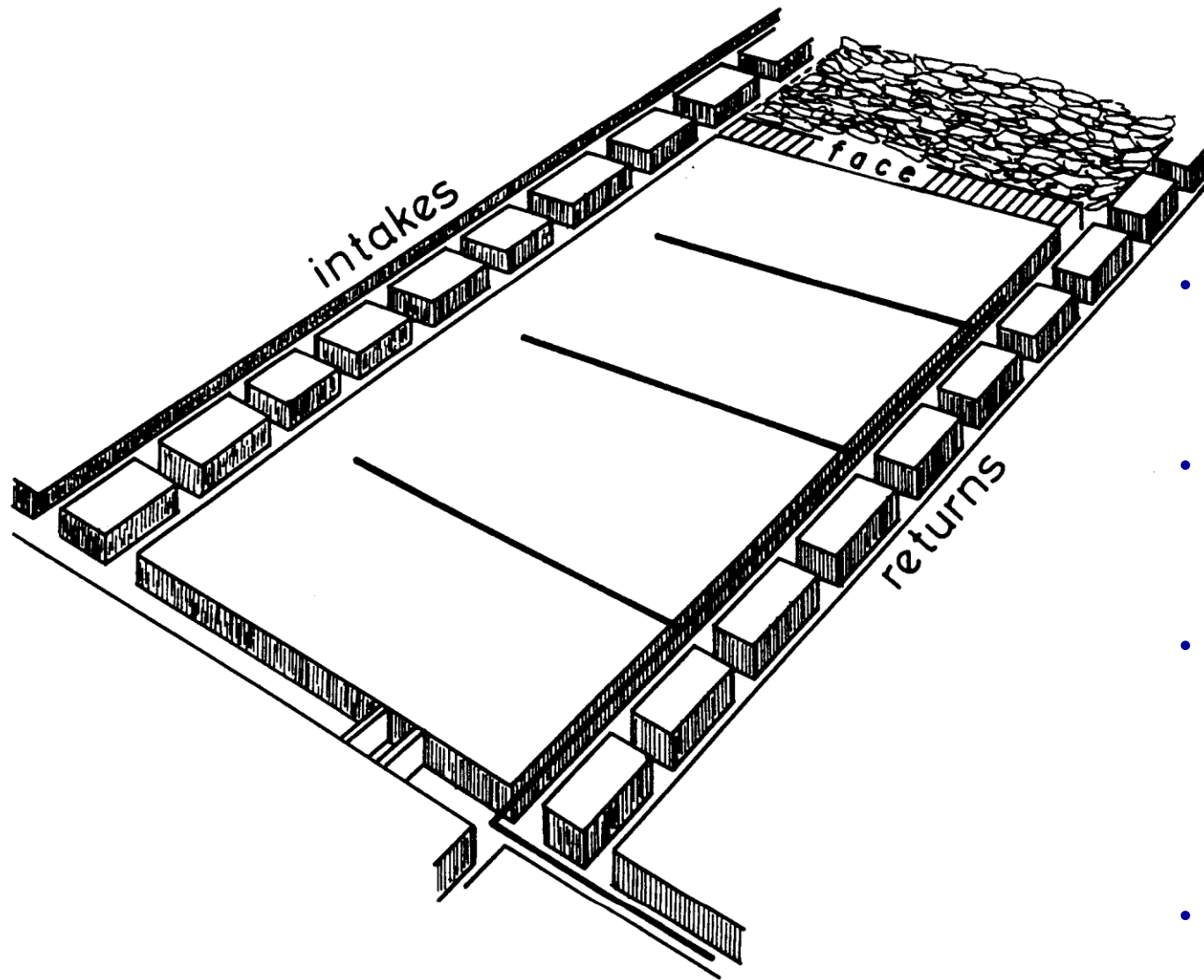
Fig. Schematic of Pre-Mine Drainage from Lateral Wells Drilled from the Surface



Layout of a horizontal borehole methane drainage system showing both in-mine and surface facilities

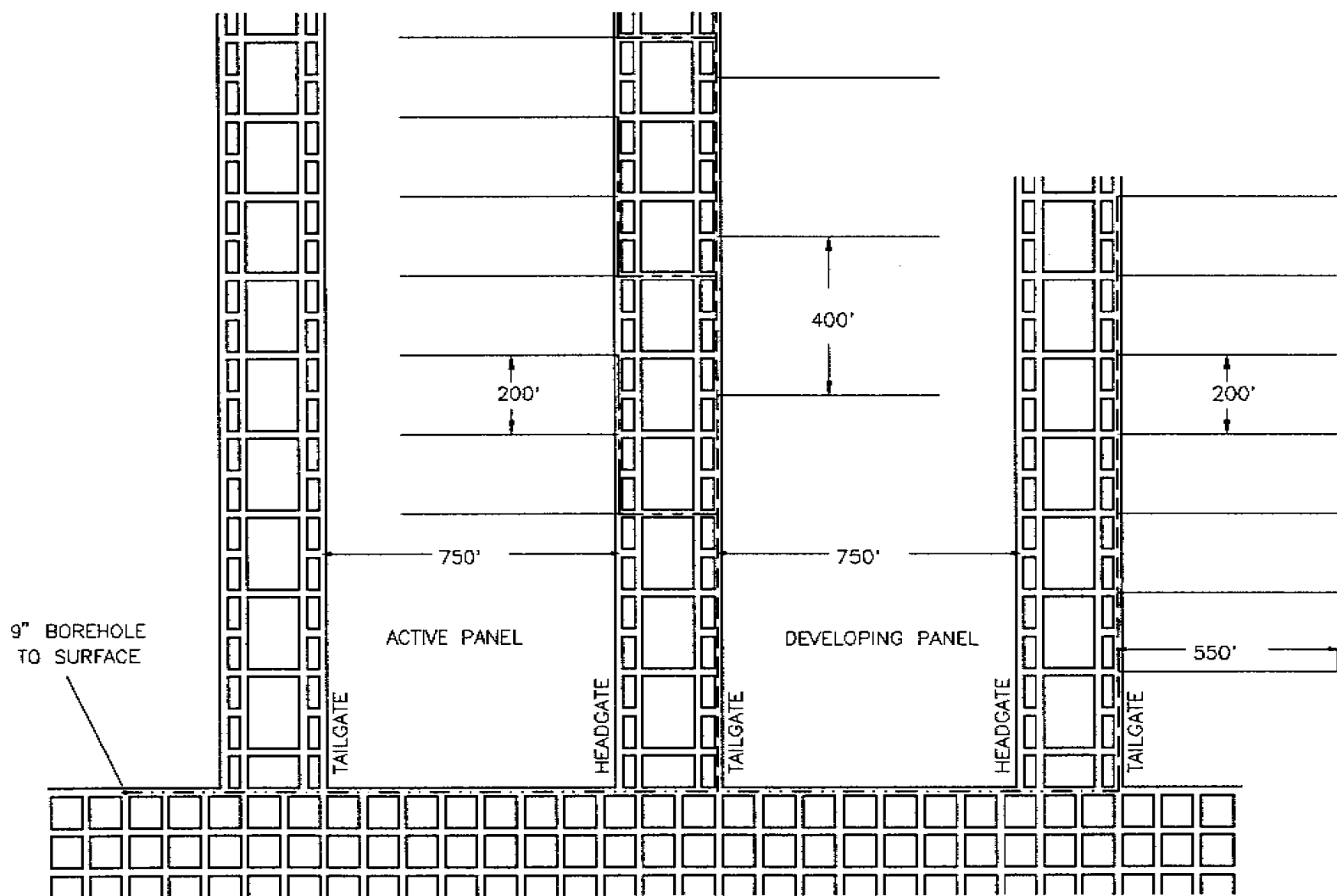
In-seam drainage from seams worked by the retreating longwall systems

- Boreholes are drilled into the seam from return airway and connected into the methane drainage pipe system.
- Preferred spacing of the holes depends upon the permeability of the seam and may vary from 10 to over 80 m.
- The distance from the end of each borehole and the opposite airway should be about $\frac{1}{2}$ the spacing between the holes.
- Application of suction on the boreholes may be required for coals of marginal permeability or to increase the zone of influence of each borehole.
- Time allowed for drainage should be at least 6 months and, preferably, over 1 year.
- Hence, the holes should be drilled during the development of the tailgate of the longwall.



- Holes are drilled in return side during development of tailgate.
- Spacing between holes: 10-80 m depending upon permeability of seam.
- Distance from the end of each borehole and opposite airway: about $\frac{1}{2}$ the spacing between the holes.
- Suction pressure is applied to increase zone of influence of holes

Fig. In-seam gas drainage from permeable seams worked by retreating longwall system



8" PIPELINE	----
6" PIPELINE	----
2" PIPELINE	----
HORIZONTAL DEGAS WELL	----

Typical horizontal borehole pattern and pipeline arrangement at a longwall mine

The flow rate of gas from a gas drainage borehole varies with time.

- Initially high flow occurs due to expansion and desorption of gas in the immediate vicinity of the hole which may diminish fairly rapidly.
- As the zone of influence is dewatered, the relative permeability of the coal to gas increases and hence increases the gas flow.
- This is again followed by a decay as the zone of influence is depleted of gas.

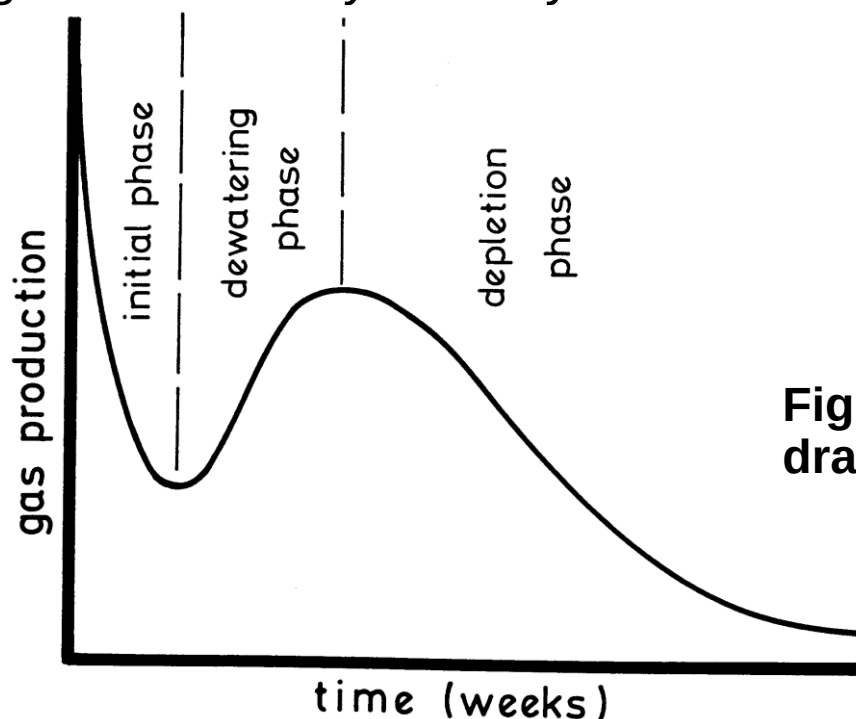


Figure: A typical life cycle of gas drainage from an in-seam borehole.

Cross-measure borehole method

- It is the most common method used for methane drainage.
- Cross-measure boreholes either upwards or downwards are drilled from roadways (usually tail gate of an advancing longwall face) in a working seam.
- Upward holes are more common because methane usually accumulates in the bed-separation cavities in the roof of the seam and is driven out of these cavities as the roof breaks and collapse.
- The holes are inclined depending on the dip of the strata.
- In flat seams the inclination is generally at an angle of 50-60° from the horizontal over the waste.
- An inclination less than 45° produces less gas, causes greater dilution of the gas (because of the larger leakage of air into the hole) and increases the chances of jamming of the bore-rods in the hole.
- Sometimes good results have been claimed with holes inclined towards the face.

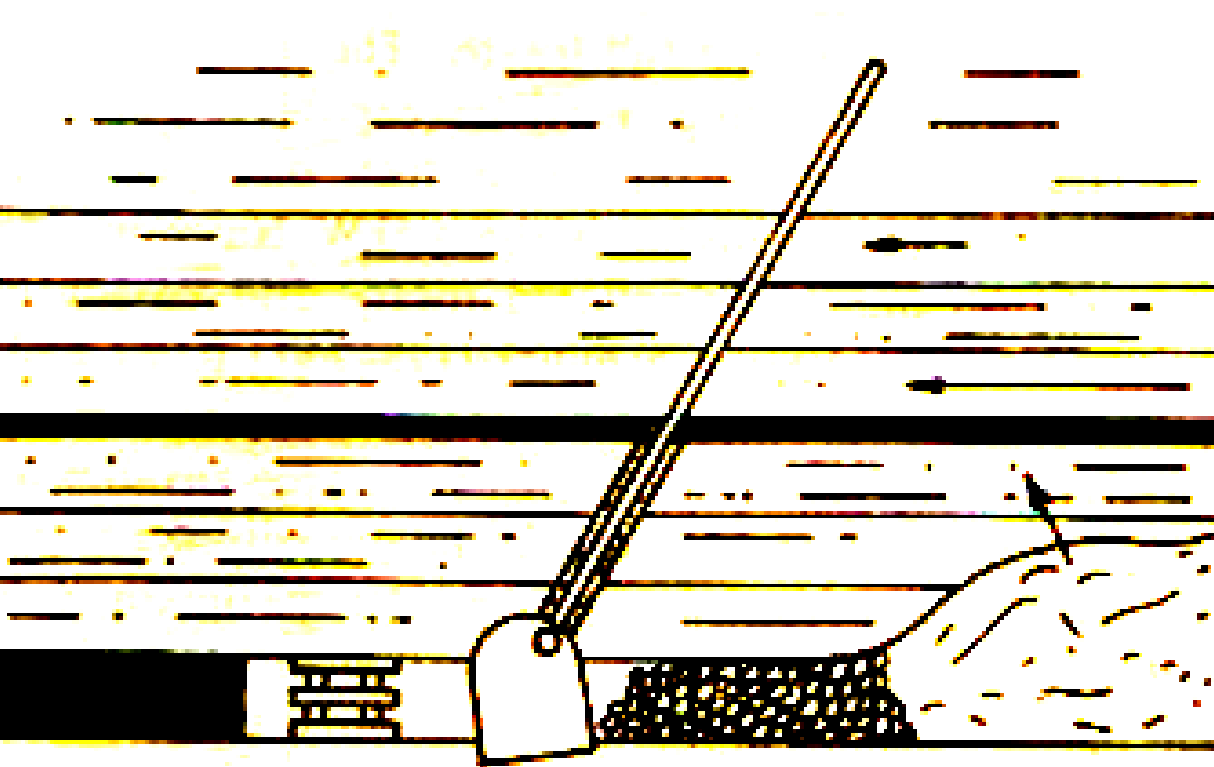


Fig. 1.2 Cross-measure borehole method of methane drainage.

- Holes are drilled in tailgate
- Length of holes: 15-100 m (typically 30-45 m)
- Dia. of holes: 65-90 mm
- Inclination of holes: 50-60 deg.
- Spacing of holes: 20-30 m
- Suction pressure applied: 1000-1500 Pa

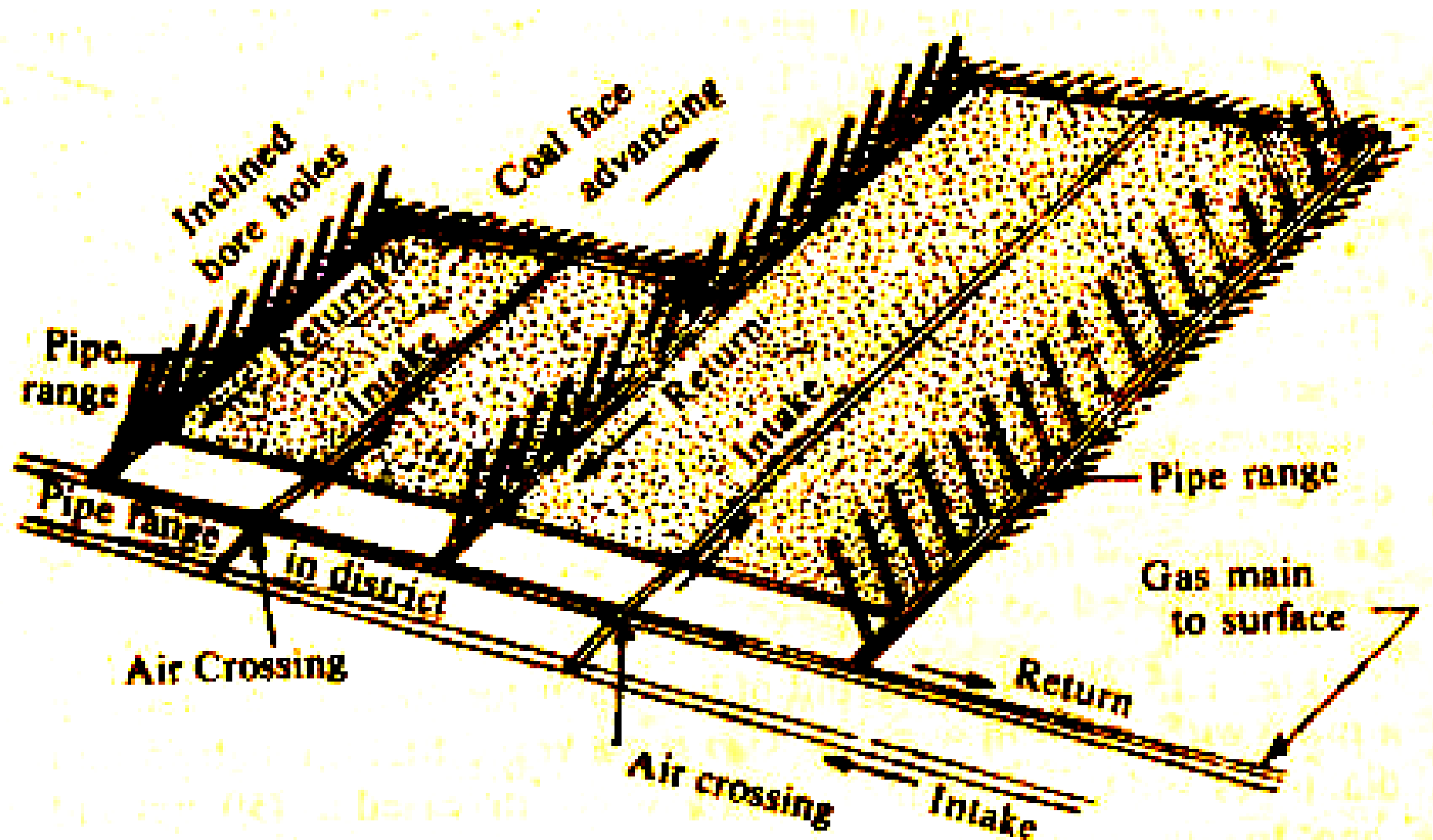


Fig. 1.15. Methane drainage (underground pipe system at a long wall face and layout of bore holes)

Gas holder

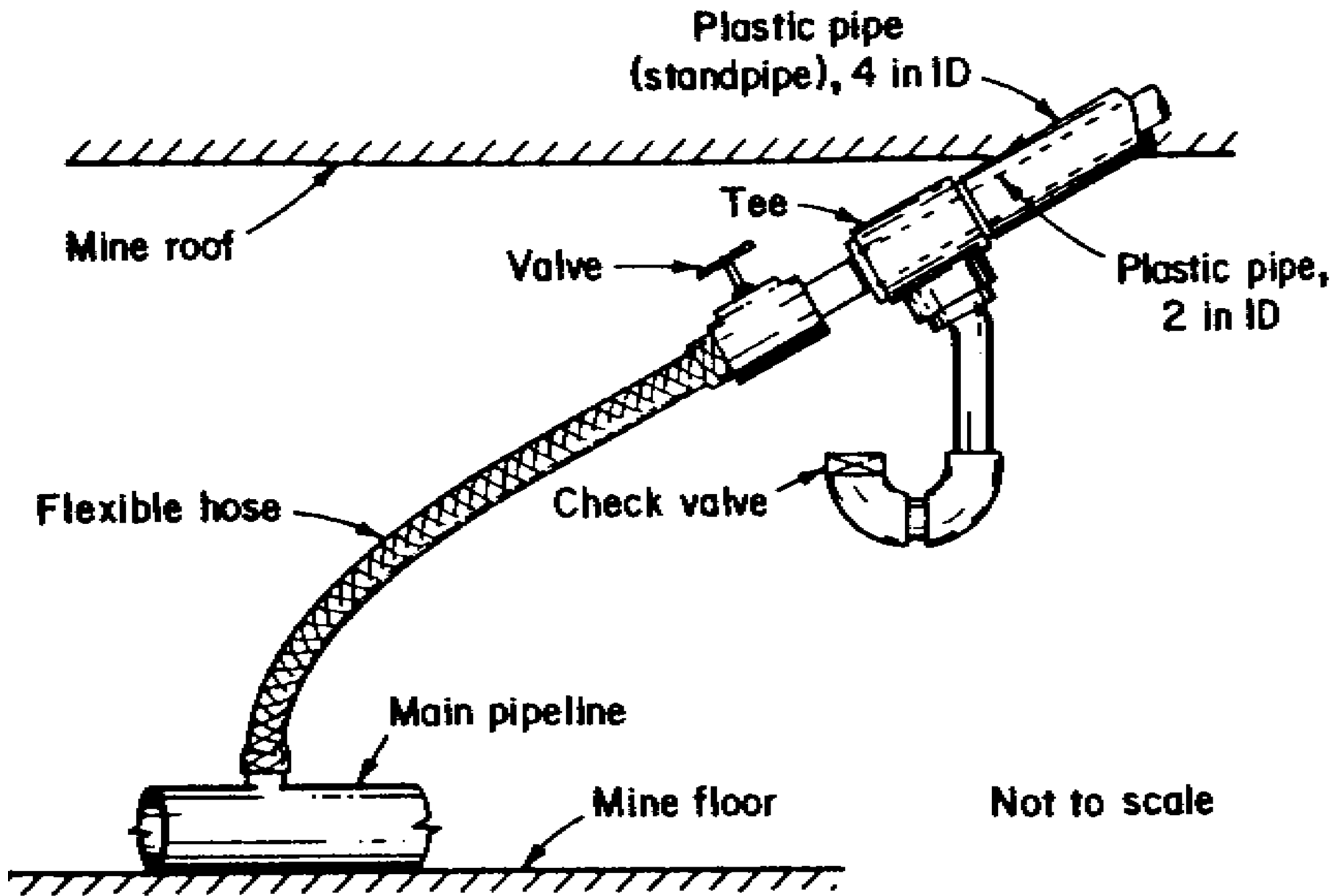


Diagram indicating angular orientation and pipeline connection for a cross-measure borehole

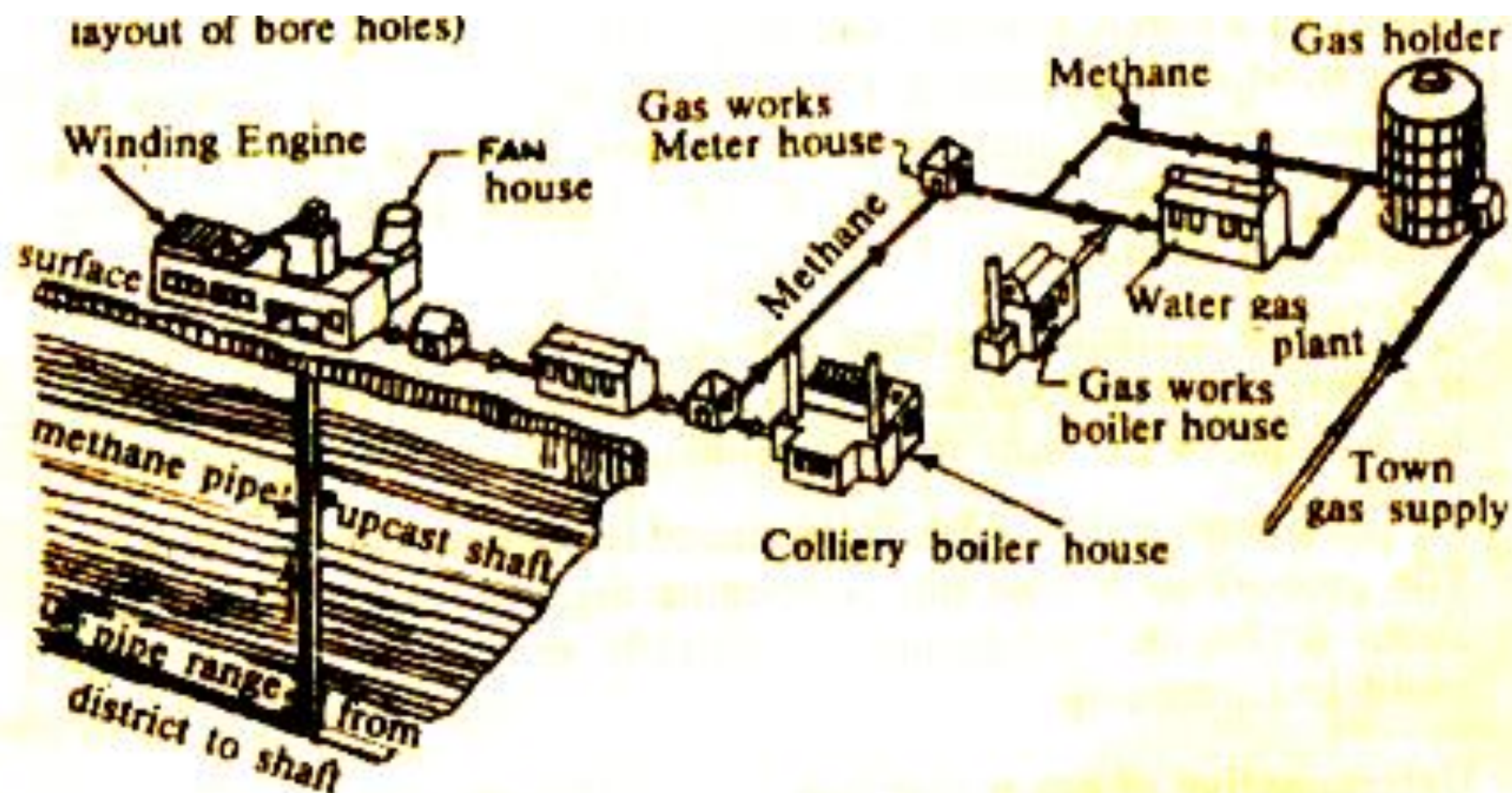


Fig. 1.16.,Methane drainage; schematic layout at the surface

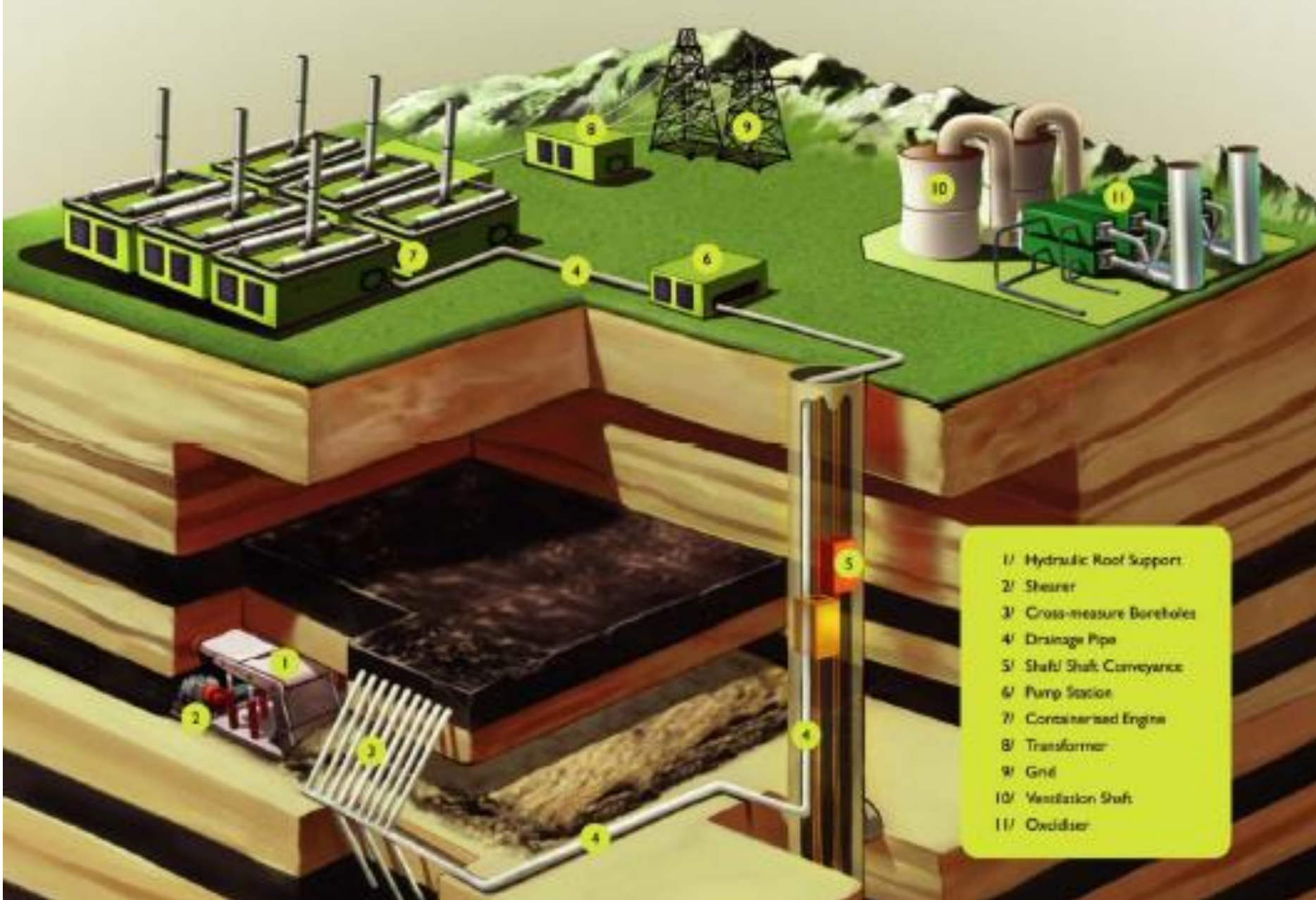


Fig. Schematic of an Underground Coal Mine Drainage System and Surface Facilities for Energy Recovery and Abatement of CMM

- Normally boreholes are placed in the return airway, since
 - the ventilating pressure drives the gas in the goaf towards the return airway.
 - the haulage in the intake makes it difficult for the bulky drill to operate due to lack of space.
- Boreholes are usually 65-90 mm in dia. and spaced at intervals of 20-30 m along the road so that they are not too close to each other to be very costly nor they are too far apart for adequate drainage.
- Their length varies from 15 to 100 m (commonly 35-40 m) depending upon the adjacent gas-bearing seams which have to be intersected for drainage.
- Holes are started off with a dia. of 115 mm and collar pipes 10m long and 90 mm in dia. are connected to them before they are drilled to the full length.

- The holes for CH₄ drainage need special drilling machines capable of drilling long holes of large diameter cheaply.
- One such machine which found suitable is the 4.5 kW Nusse and Graffer Fortschritt PIV/6 pneumatic-powered rotary drill having a rotational speed of 125 to 250 rpm and using multi-point fir-tree type of bits.
- Usually the rate of gas emission rises for the first two weeks until a maximum is reached at a distance of 30 to 300 m from the face, after which the rate of emission falls.
- Usually a suction pressure of 1000 to 1500 Pa is used to drain the gas.
- Suction should be high enough to overcome the friction of the pipe ranges, but, at the same time, should not be so high that air may leak into the goaf and dilute the methane.

Advantages of cross-measure borehole method :

- CH₄ discharged into the ventilation system is decreased by 50-60% or sometimes more due to this method.
- Of all the methods, this method is the simplest and cheapest.

Superjacent Heading or Hirschbach Method

- This method is applicable where an unworkable seam exists some 25 to 35 m above the seam being worked.
- In this method, headings with cross-sectional area of 5-7 m² are driven at a height of about 25-30 m on top of the working seam.
- They are so positioned that their vertical projections lie midway between the gate roads in the working seams.
- It is better to locate the drive in a coal seam at least 0.3-0.4 m in thickness if such a seam is available. Otherwise, they may be driven in stone.
- For economy, an already existing drift or one driven for some other purpose may be chosen for methane drainage.
- Both along-the-seam and cross-measure boreholes are drilled from these headings.
- The headings are sealed at the outbye end by dams through which pipes are left.
- Methane is drawn from the headings through these pipes by applying a suction of 203 kPa using vacuum pump.

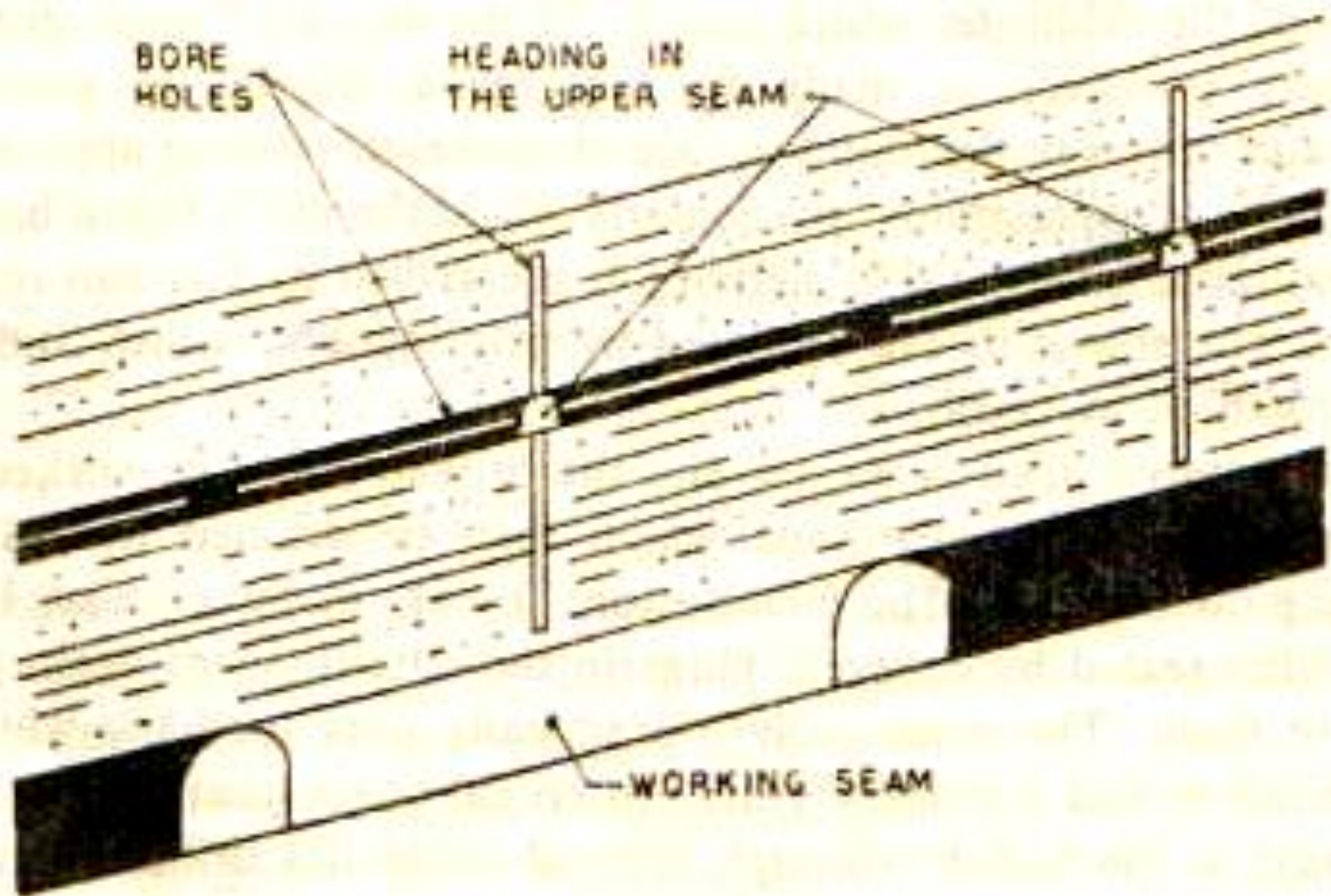
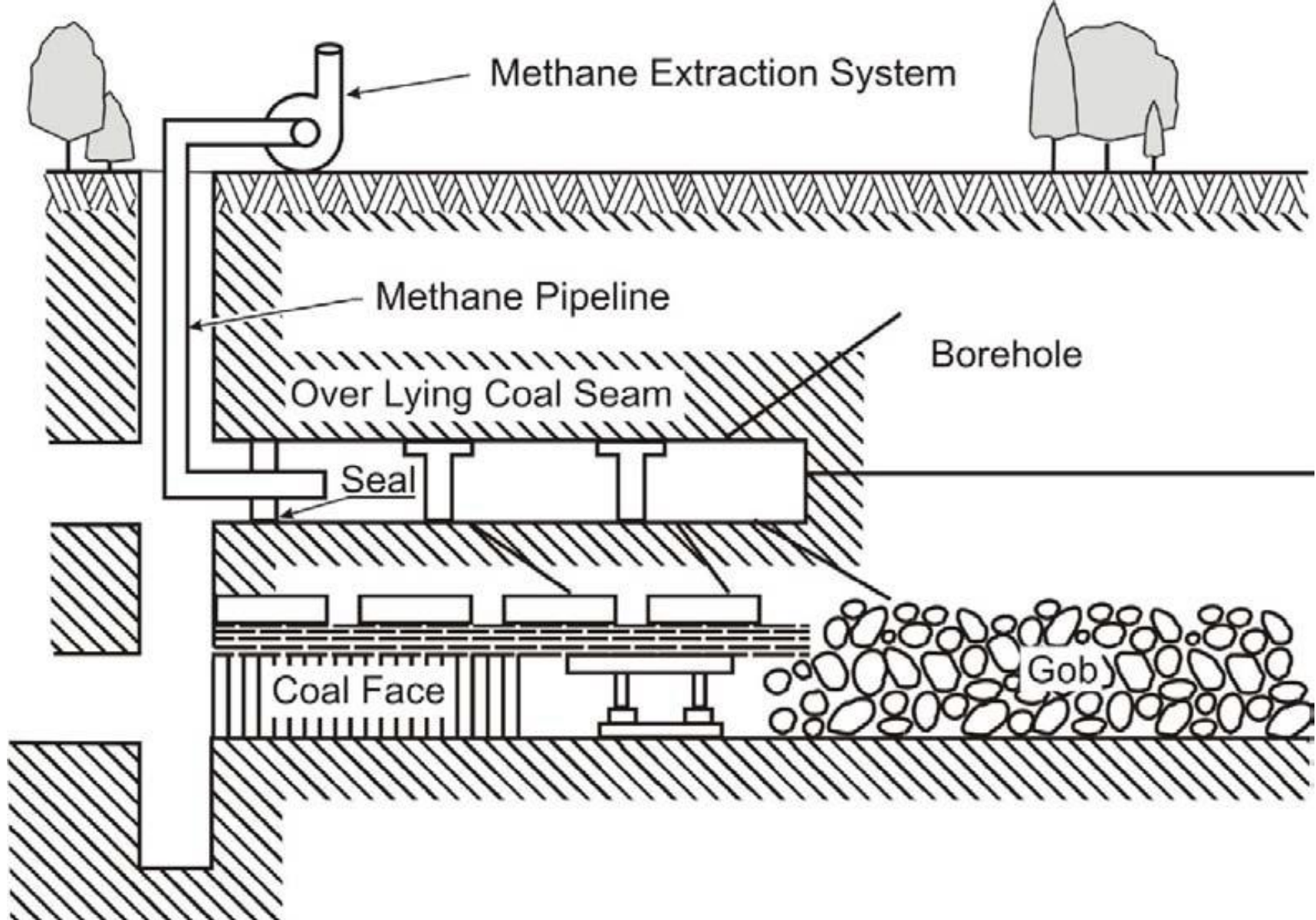


Fig. 1.3 Hirschbach method of methane drainage.



Methane drainage by the superjacent method

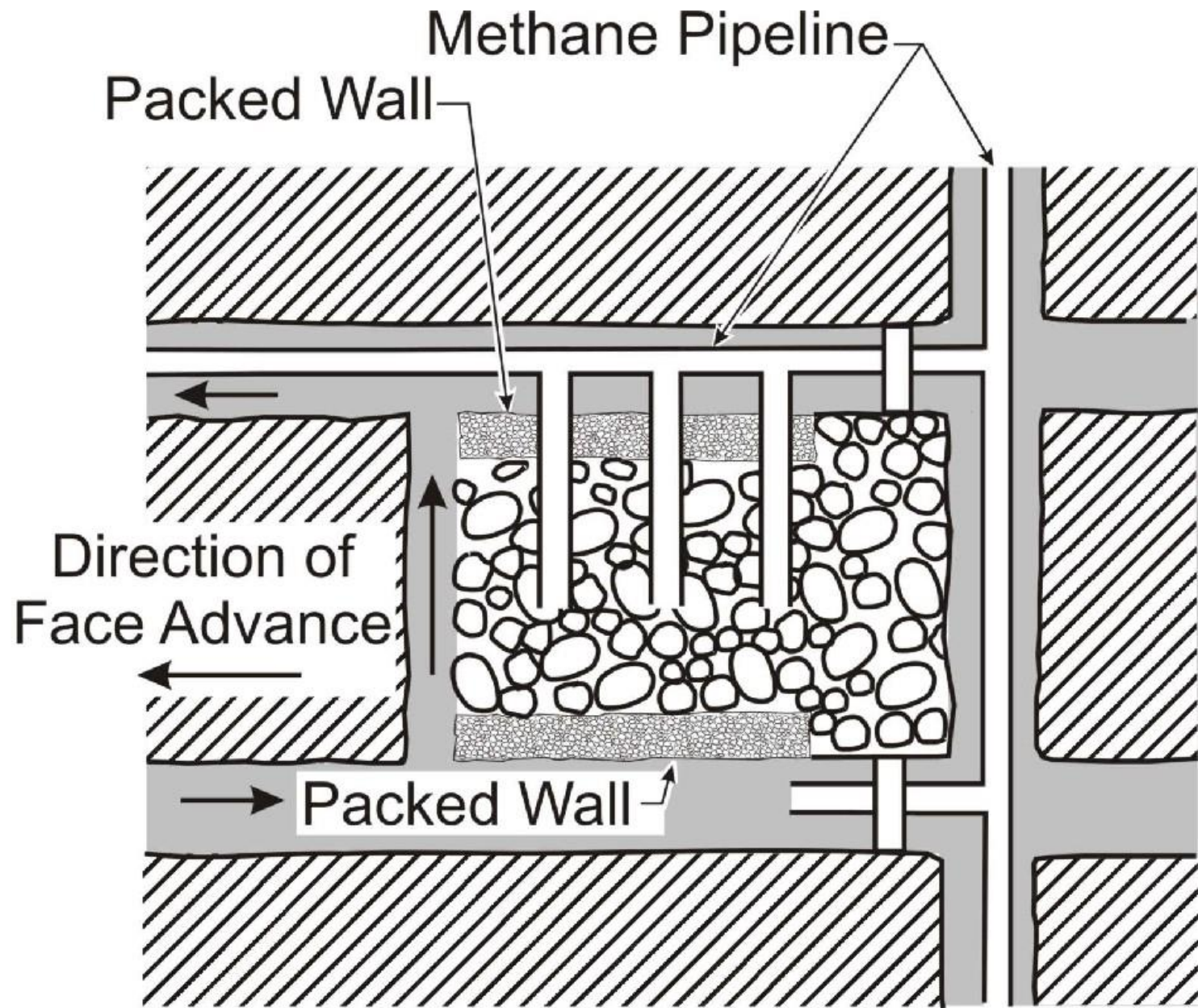
- This method was used mainly for retreating longwall faces in highly gassy seams in French mines.
- This method is prevalent in mines in Eastern Europe and China.

Advantages and disadvantages of Hirschbach method

- This method yields maximum quantity of gas with a high methane percentage.
- The driving and drilling operations are done outside the working seam and hence do not interfere with normal mining operations.
- This is a suitable method for retreating longwall and bord-and-pillar workings.
- This method is not very successful when the seam being worked has a massive sandstone cap because enough roof fissures are not developed with such a roof and therefore gas flows more easily to the face than to the drainage drift.

Pack-cavity/corridor method

- This technique is used mainly in Russian coal mines.
- In this method corridors or webs parallel to the face are left in solid pack using packed walls, partial or complete stowing, and plastic sheets or massive stoppings at intervals of about 40 m and within 20 m from the intake-and-return-gate roads.
- Individual pack cavities are connected to a network of pipeline and main pipe range of 150-300 mm in dia. in return airway through pressure gauges, orifice type flow meters, water drain taps, sampling taps and valves etc.
- The gas is collected by applying a suction head of 250 to 350 Pa using vacuum pump.



Methane drainage by the packed cavity method

Disadvantages:

- This method yields minimum quantity of gas which is often much diluted.
- In this method, only a closely controlled small suction head can be applied to prevent air leakage into the goaf causing
 - spontaneous heating and
 - dilution of the gas drained.

Gob drainage by surface boreholes

- Methane accumulates at high conc. in the voids or goaves of the longwall panel and called 'gob gas'.
- If this gas is not removed, then it will migrate towards the working horizon and become a load on the ventilation system of the mine.
- Capture of this "gob gas" is accomplished underground either by cross-measures drainage or by drilling vertical boreholes from the surface.
- This method is commonly used in longwall mining in the United States, where coal seams are generally thin, shallow, and relatively more permeable.

- Typically, 3 or 4 holes are drilled from surface rigs at intervals of 500 to 600 m along the centreline of the panel and ahead of the coal face.
- The holes may be 200-250 mm in dia. and drilled to within 8-10 m of the top of the coal seam .
- The holes should be cased from the surface to a depth that is dictated by the local geology.
- A perforated liner can be employed in the rest of the hole to inhibit closure from lateral shear.
- The initial gas made from the surface holes is likely to be small.
- However, as the face passes under each borehole, the CH₄ that accumulates in the caved area will be drawn towards that borehole.

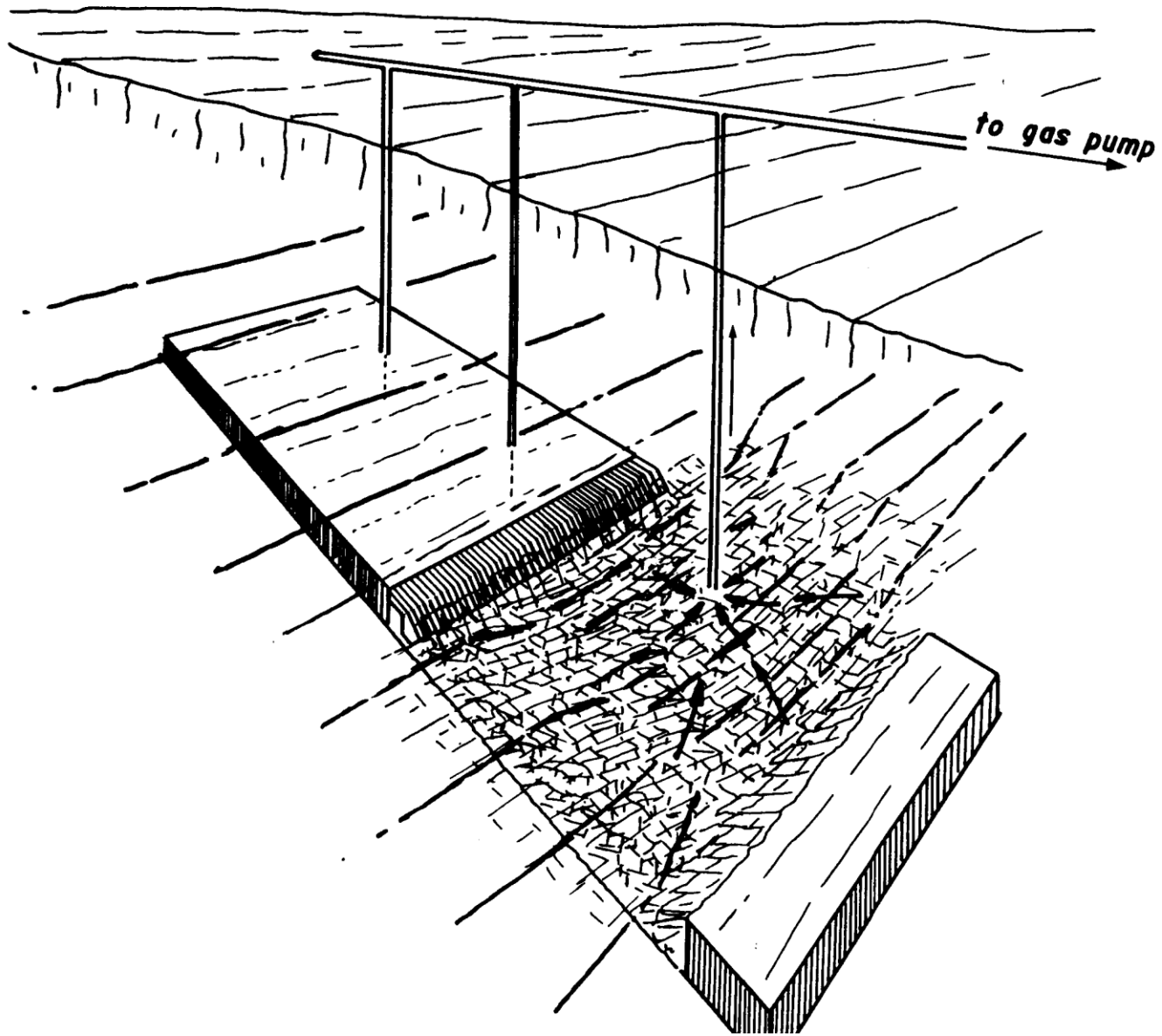
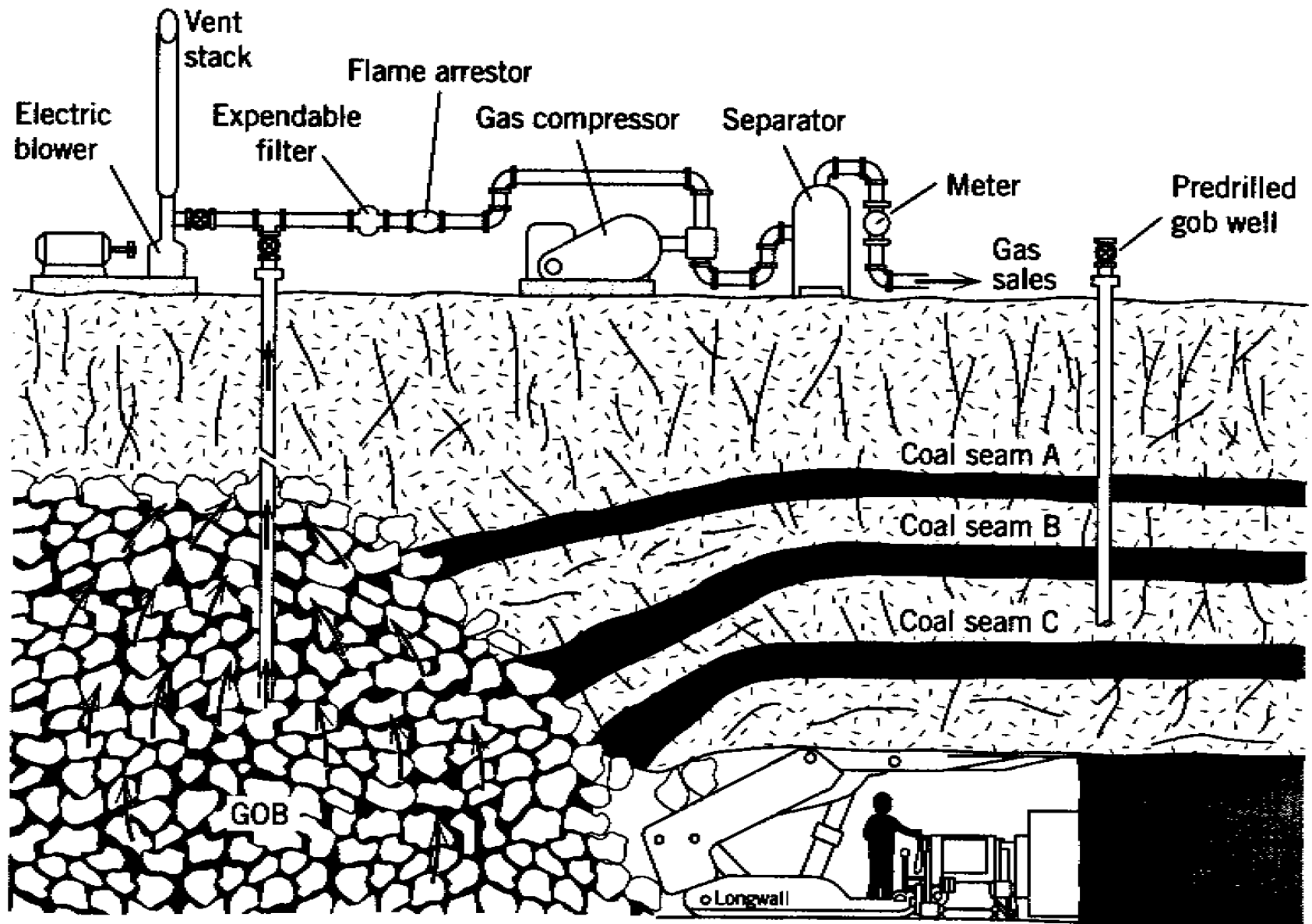


Fig. Gob drainage of a longwall panel



Gob well setup above an active longwall section

- Bed separation assists drainage from the complete gob area.
- The first borehole should be located far enough from the face start line (typically about 150 m) of longwall face to ensure that it connects into the caved zone.
- When a hole becomes active, the rate of gas production increases sharply and may yield over 50,000 m³/day of commercial quality CH₄ for a period of several months, depending upon the rate of mining.
- Gas drainage pumps located on surface ensure that the gas flow remains in the correct direction and may be employed to control both the rate of flow and gas purity.
- If the applied suction is too great, then ventilating air will be drawn into the gob and may cause excessive dilution of the drained methane.
- In addition to longwall panels, gob drainage by surface boreholes can be utilized in pillar extraction areas.
- This technique can result in very significant reductions in emissions of methane into mine workings.

Methane capture ratios achieved in practice for post-mining methane drainage techniques

Methane drainage technique and methane capture ratios	Remarks
Packed cavity method (after Lidin [1961]):	
20%–40%	Caving longwalls.
30%–50%	Partially stowed longwalls.
60%–80%	Fully stowed gobs.
Cross-measure boreholes (after Kimmins [1971]):	
59%–70%	Highly gassy mines with specific emissions 3,000–6,000 ft ³ per ton.
Superjacent method:	
50%	For multiple coal seams in the gas emission space.
Vertical gob wells:	
30%–80%	The methane capture efficiency depends on the number of gob wells per longwall panel and production techniques.

DETECTION OF METHANE

Many firedamp detectors utilizes various physical properties of Methane-air mixture e.g.:

- Density
- Refractive Index
- Thermal conductivity
- Inflammability
- Change of volume of combustion
- Diffusion
- Absorption of infra-red rays etc.

Methods of Detection of Firedamp

- Detection by Flame Safety Lamp
- Detection by using Methanometer

Methanometers:

Two types:

1. Automatic types: Gives a signal when the gas % reaches a certain predetermined value
2. Estimating types

MSA D6 Methanometer

Principle of Working

Based on the principle of Wheatstone Bridge which utilizes the change in resistance of a wire on heating by combustion of CH_4

Constructional Features:

- A balanced Wheatstone Bridge circuit
- A sensitive galvanometer and
- A battery

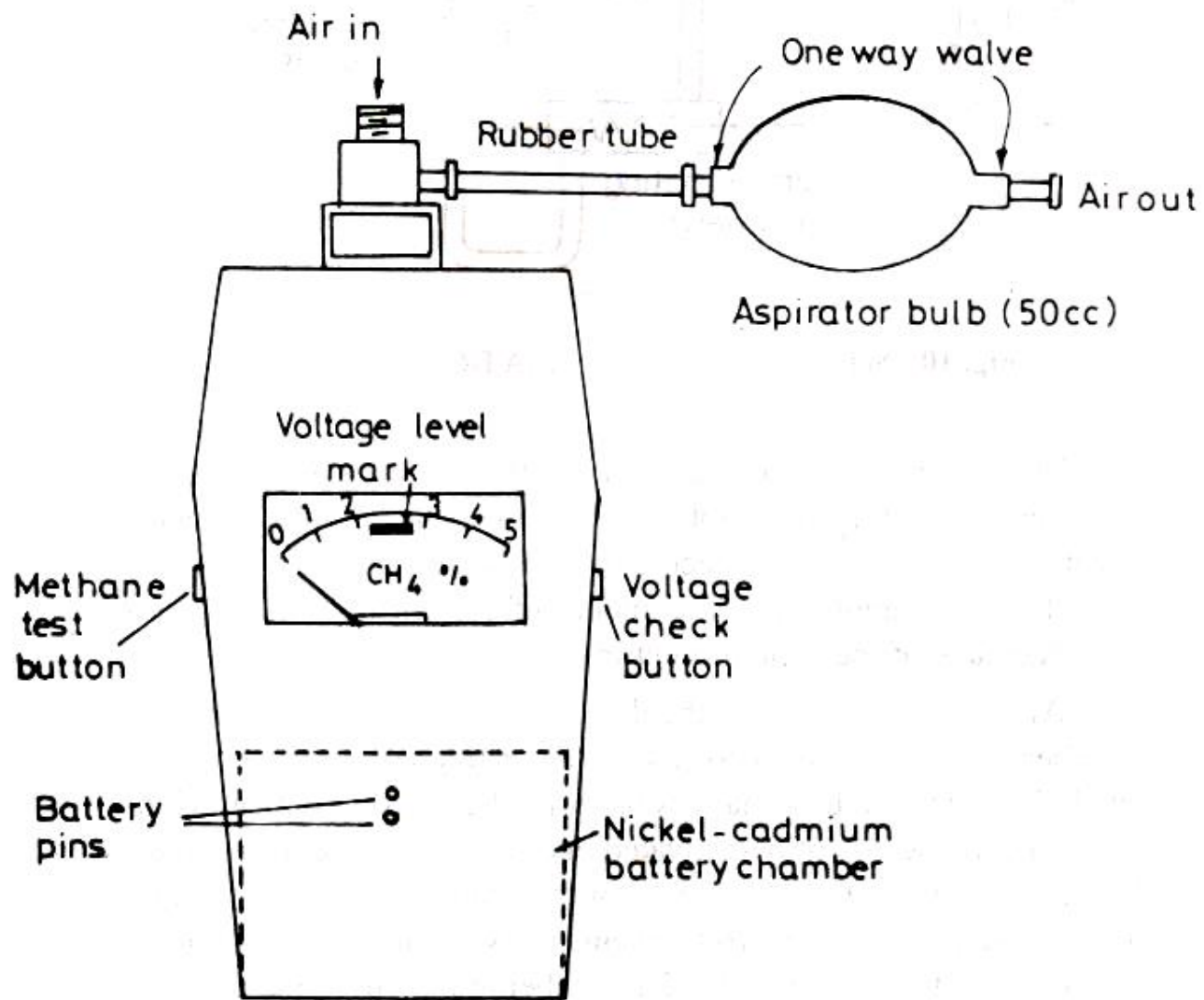


Fig. 9 : MSA D6 Methanometer

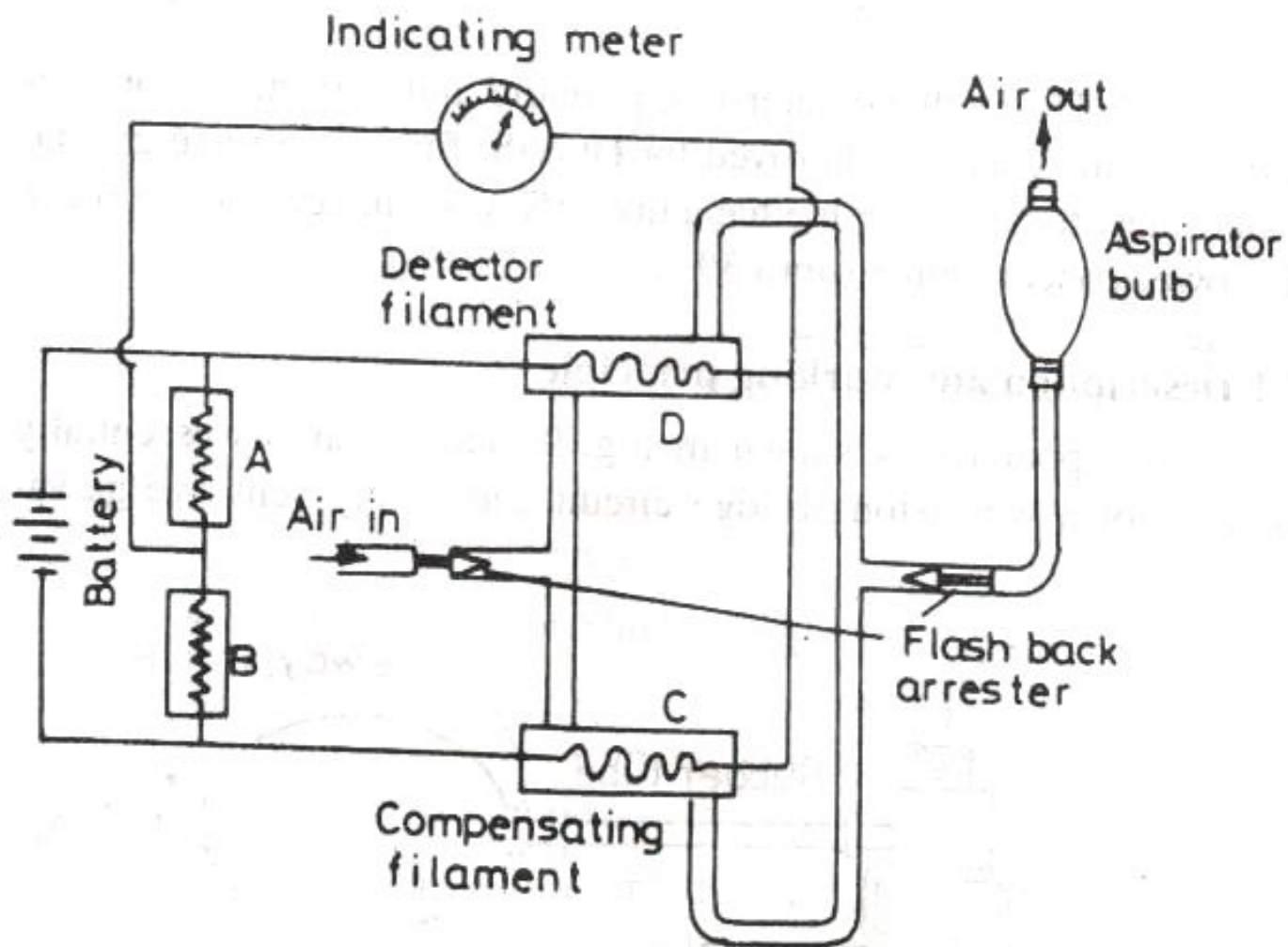


Fig. 10 : Schematic Diagram of MSA D6 methanometer

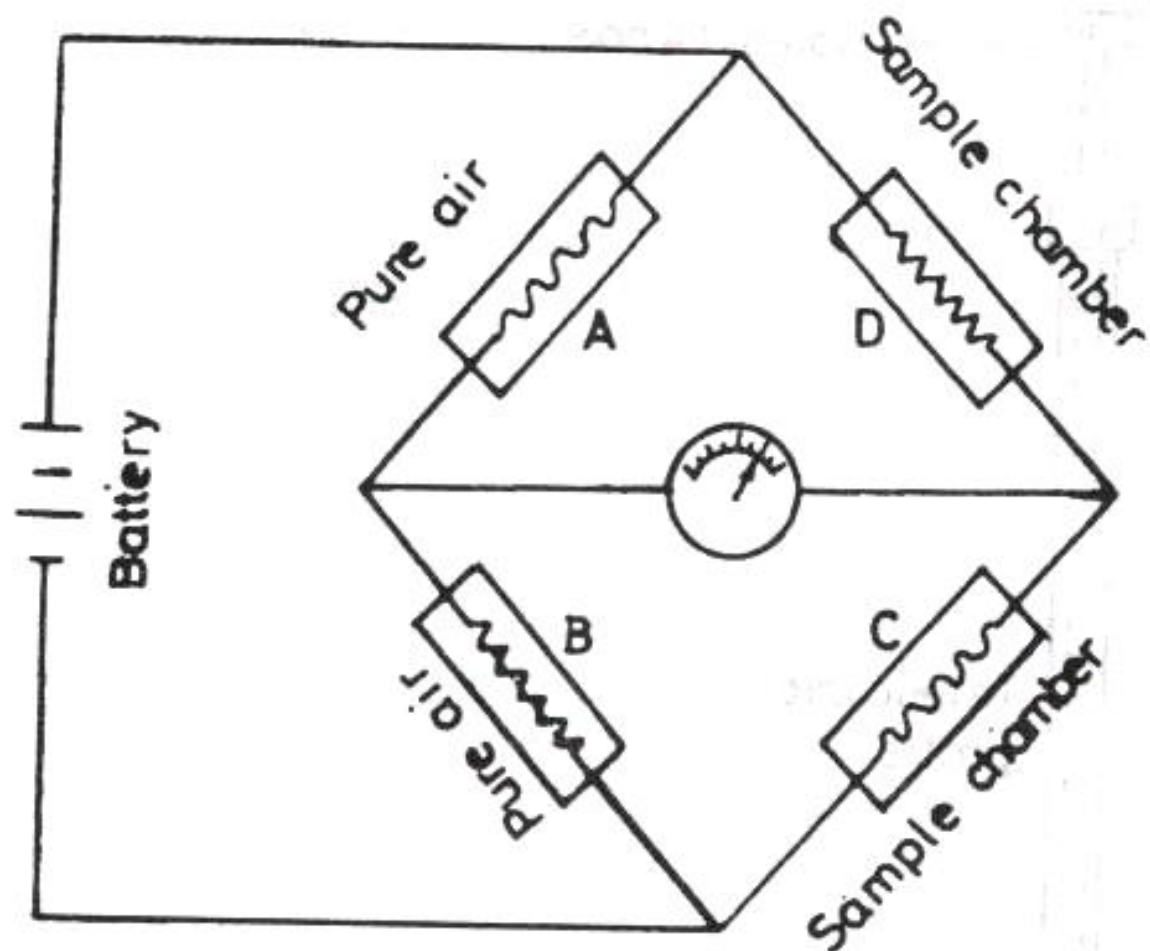


Fig. 11 : Wheatstone bridge circuit of MSA D6 methanometer

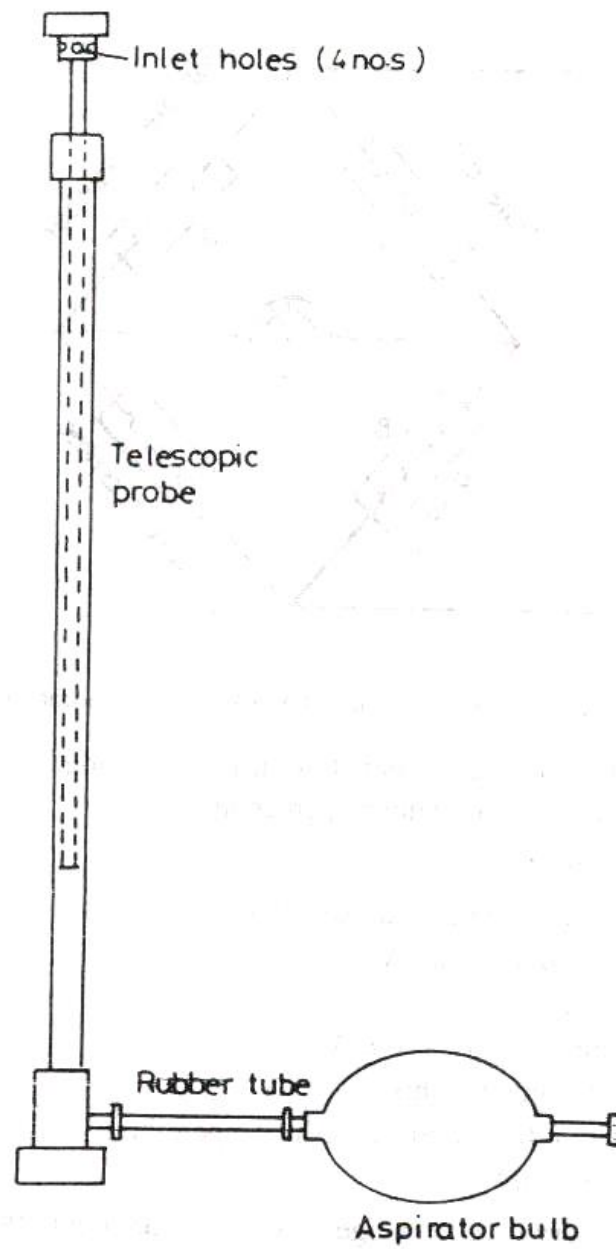


Fig. 12 : Telescopic probe for MSA D6

SAMPLING AND ANALYSIS OF MINE AIR

Sampling of Mine Air

Collection of Mine Air Samples

- **Spot samples:** Collected at predetermined places in mine airways for routine analysis.
- **Where variation in composition is expected over the cross-section of the airway:** inlet of the sample holder may be traversed over the section of the airway in a regular fashion.
- **Where distinct variation in composition exists:** as in CH₄ layering, spot samples at several intervals have to be taken at the airway cross-section.

Methods of air sample collection:

- **Water or salt-solution displacement:**
 - Air samples are collected by emptying out the liquid contained in the sample holder.
 - Water displacement is unsuitable when CO₂ is to be accurately determined since CO₂ is readily soluble in water.
 - A 22% aqueous solution of common salt is suitable for this case.
- **Air displacement:** by sucking through mouth or by aspiration through a rubber aspirator or a hand or foot-operated pump.
 - Sufficient air should be drawn through the sample holder to fully flush the holder.
- **In evacuated sample holders:** in the form of glass bulbs evacuated by vacuum pump.

Sample holders

Gas samples holders both displacement and vacuum type are generally of 250-500 cm³ capacity.

- **Glass:** is the most suitable material for sample holder.
- It is fragile and needs careful handling

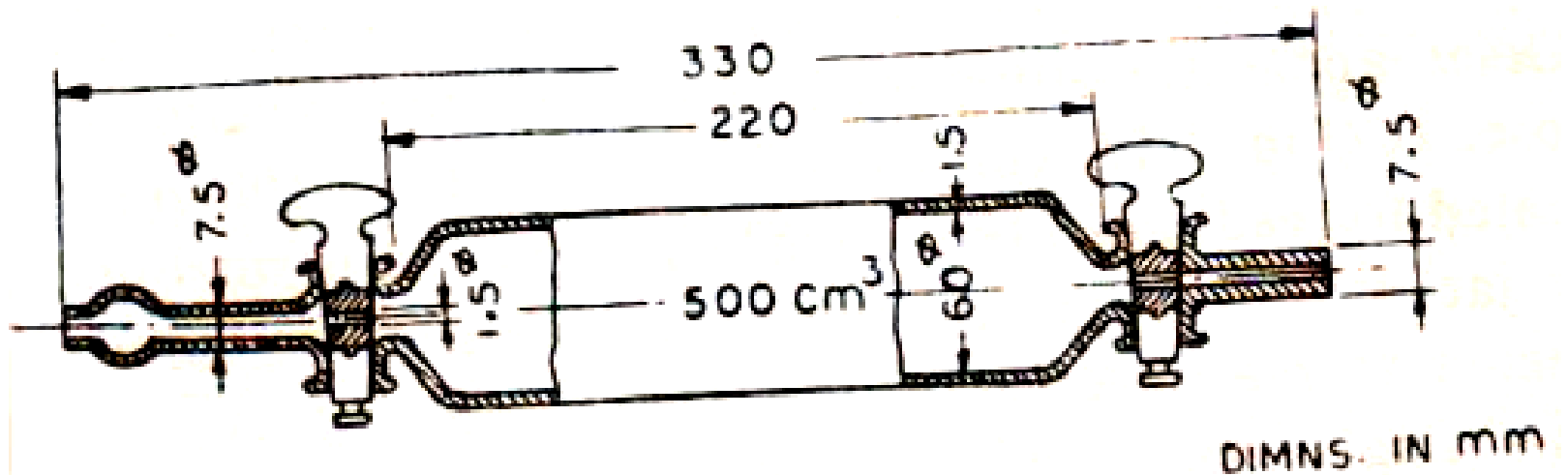


Fig. 1.16 Displacement-type glass sample holder.

Metallic holders:

- Stop valves tend to leak if not properly fitted and greased
- Iron and tin are unsuitable as they readily absorb O₂ in the sample
- Brass and other non-ferrous metals can be used but if H₂S and NO₂ are present, copper and its alloys become unsuitable.

Rubber or polythene: should not be used as gases diffuse through them

Analysis of Mine Air

- By mine air analysis, normally conc. of gases like O₂, CO₂, CO, CH₄ and N₂ which are the common constituents of mine air are determined.
- Rare constituents like H₂S, SO₂ and NO₂ are estimated by accurate spot detectors.

Methods of Analysis

Chemical Analysis:

- Accurate and cheaper method
- Time consuming
- Requires large amount of gas samples

Common Chemical gas-analysis Apparatus

- Orsat Apparatus
- Haldane Apparatus: requires small samples

Quantity of air samples needed for analysis

- **For analysis in the Orsat Apparatus:**
 - 250-500 cm³ air sample is needed
- **For analysis in the Haldane Apparatus:**
 - requires very small quantity and only 70 cm³ is enough for 3 analysis

Physical Methods of Analysis

- Quicker and accurate analysis of samples.
- Require less sample.
- Instruments are of high cost.

More popular apparatus are

- Infra-red Gas Analysers
- Gas Chromatographs

Orsat Apparatus

- Used for rough and routine mine-air analysis

Constructional features:

- Burette D of 100 cc capacity graduated to read down 0.1 cc surrounded with a water jacket F.
- Connected at lower end by a rubber tubing to a bottle E containing NaCl solution.
- Two pipettes, A & B and combustion chamber C connected to the burette through the taps T1, T2 & T3 resp.
- Pipette A contains a soln. of 66 g of caustic potash (KOH) dissolved in 200 cm³ of distilled water for absorbing CO₂.
- Pipette B contains a soln. of alkaline pyrogallol (white crystalline powder of trihydroxybenzene, C₆H₆O₃): formed by dissolving 10 g of pyrogalllic acid in 100 cc of nearly saturated KOH soln. for absorbing O₂.
- The combustion chamber is filled with mercury and a platinum coil for heating purpose.

Pipette A: Soln. of 66g of KOH dissolved in 200 cc of distilled water for absorbing CO₂

Pipette B: A soln. of alkaline pyrogallol for absorbing O₂

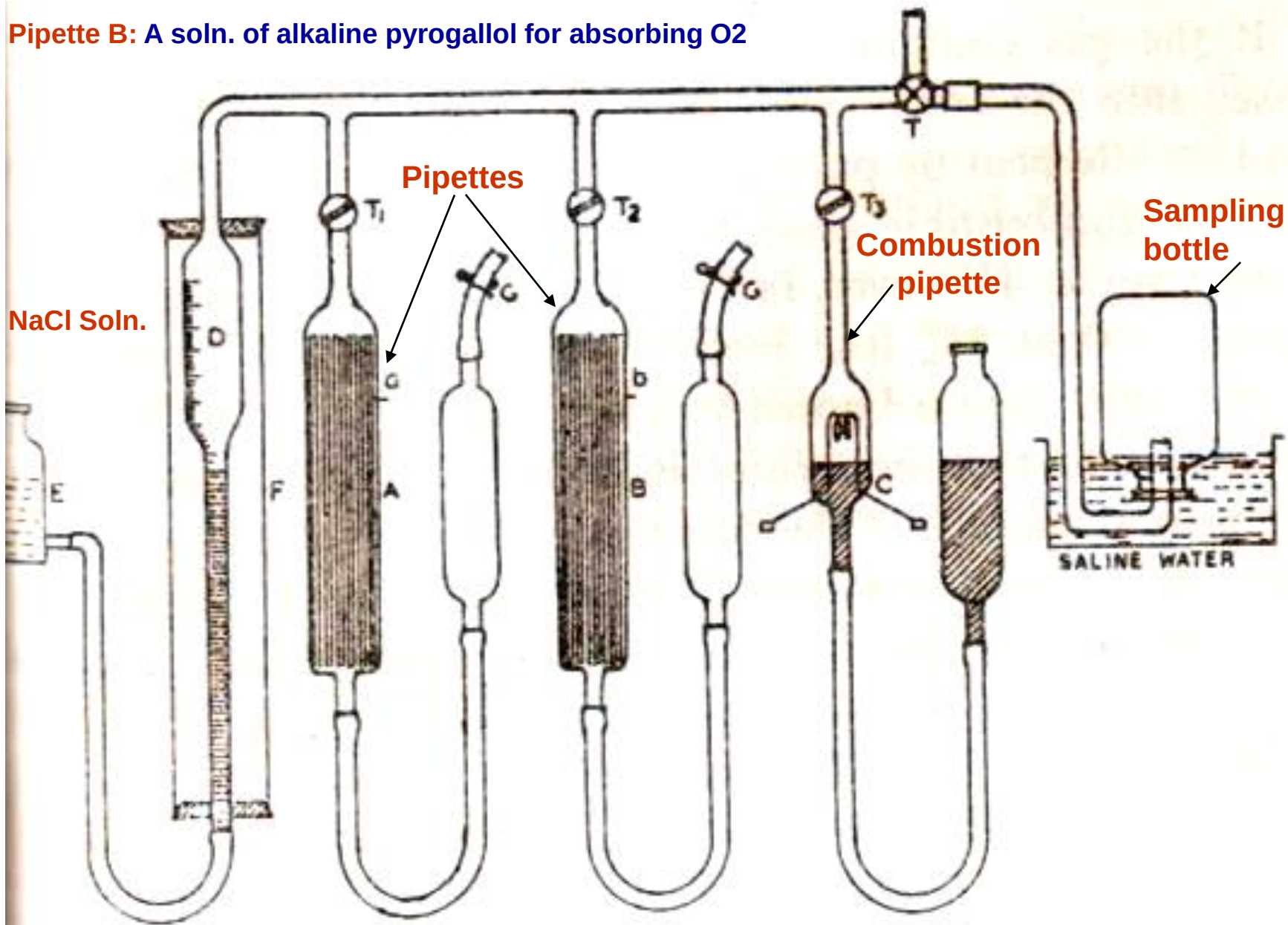
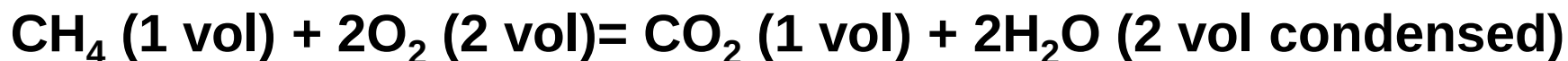


Fig. 1.17 Orsat apparatus (diagrammatic).

- Combustion of CH₄ and CO consumes O₂ and produces CO₂.

In case of CH₄ combustion:



- When 1 vol. of CH₄ burns, consumes 2 vols. of O₂ and causes a reduction in vol. equal to twice the vol. of CH₄.
- Also, combustion of 1 vol. of CH₄ produces one vol. of CO₂.

In case of CO combustion:



- Vol. of CO₂ produced after combustion is equal to the vol. of CO.

If the gas contains both CH₄ and CO

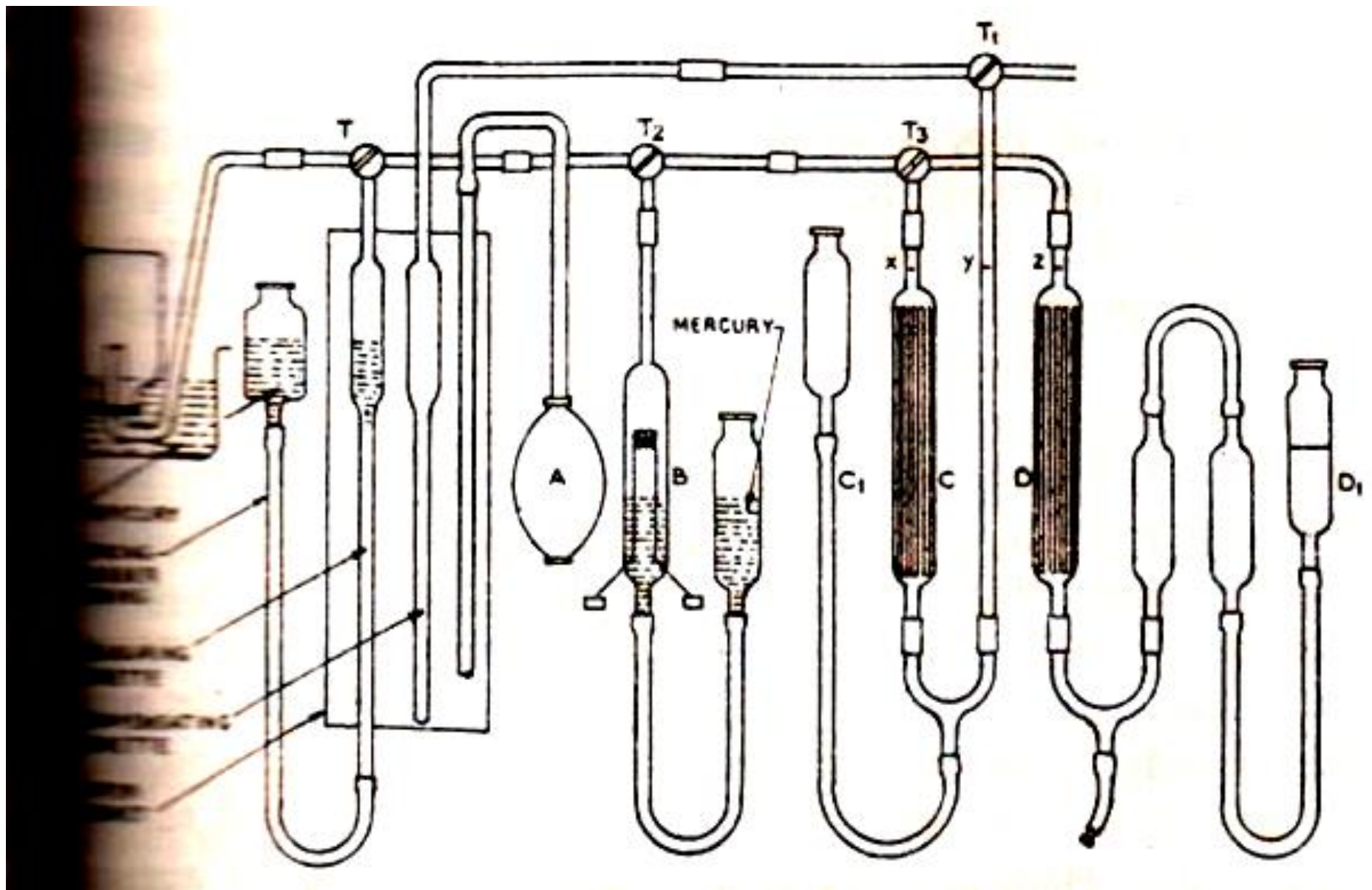
Let vol. of CH₄ = x cc

Vol. of CO = y cc

- O₂ consumed (contraction) after combustion = $2x + \frac{1}{2}y$
- CO₂ produced after consumption: $x + y$

Haldane Apparatus

- This is a more accurate apparatus for analysis of mine air.
- It is similar in principle to Orsat Apparatus.
- There are two sizes of Haldane Apparatus for different degree of accuracy:
 - 1. Small portable type** (burette capacity: 10 cc): graduated to read a min. quantity of 0.01 cc
 - 2. Large laboratory type** (burette capacity: 20 cc): graduated to read a min. quantity of 0.001 cc



Haldane Apparatus

Constructional Features:

It mainly consists of two burettes

- Measuring burette and
- Compensating burette: to take into account the variation in atmospheric conditions, viz. temp., press. & humidity.
- A water jacket covering both the burettes.
- Two pipettes:
 - one containing 36% soln. of caustic potash for absorption of CO₂
 - and the other, a soln. of alkaline pyrogallol (white crystalline powder of trihydroxybenzene, C₆H₆O₃) for absorption of O₂.
- Surface of pyrogallol soln. is exposed to atmosphere usually covered by a layer of paraffin to prevent oxidation of reagent.

Infra-red Gas Analyzers

- These are fairly extensively used now a days for routine analysis of CH₄ and other mine gases.
- The method is very quick and has an accuracy of the order 0.1%.

Principle of Working:

- It utilizes the principle that a particular gas absorbs radiation of a particular wave length, e.g. CO₂ absorbing radiations of 4.2 μm , CO of 4.66 μm and CH₄ of 7.5 μm .
- When an incident beam of this particular wave length passes through a cell containing the gas, it is partly extinguished.

- The amount of extinction is governed by Beer's law

$$I = I_0 e^{-aLc}$$

Where

I_0 = intensity of incident beam

I = intensity of transmitted beam

a = absorptivity index or absorption coefficient (absorptive power)

L = optical path or length of the cell

c = molar concentration (kmol m^{-3})

Constructional Features:

Infra-red spectrometer essentially consists of

- A **source of infra-red radiation** of sufficient range of wave length so as to cover all the gases present in the sample.
- A **slit and a collimator** which produce a narrow beam of parallel rays.
- A **prism** which disperses the beam to its components of various wave lengths.
- An **adsorption cell** for holding the sample and
- **The receiver** which records the intensity of transmitted radiation.

Working

- By rotating the prism a beam of a particular wave length is made to traverse the adsorption cell which have windows of suitable material that does not absorb radiation.
- For wave lengths between 3-9 μm fluorite windows are used.
- For wave lengths of 8-16 μm and 15-20 μm respectively, rock salt and sylvine (a colorless vitreous potassium chloride mineral) are used.
- The prism is also made of same material as the windows.
- The receivers can be either bolometers or thermocouples used in conjunction with suitable amplifiers to produce measurable voltages.

GAS CHROMATOGRAPH

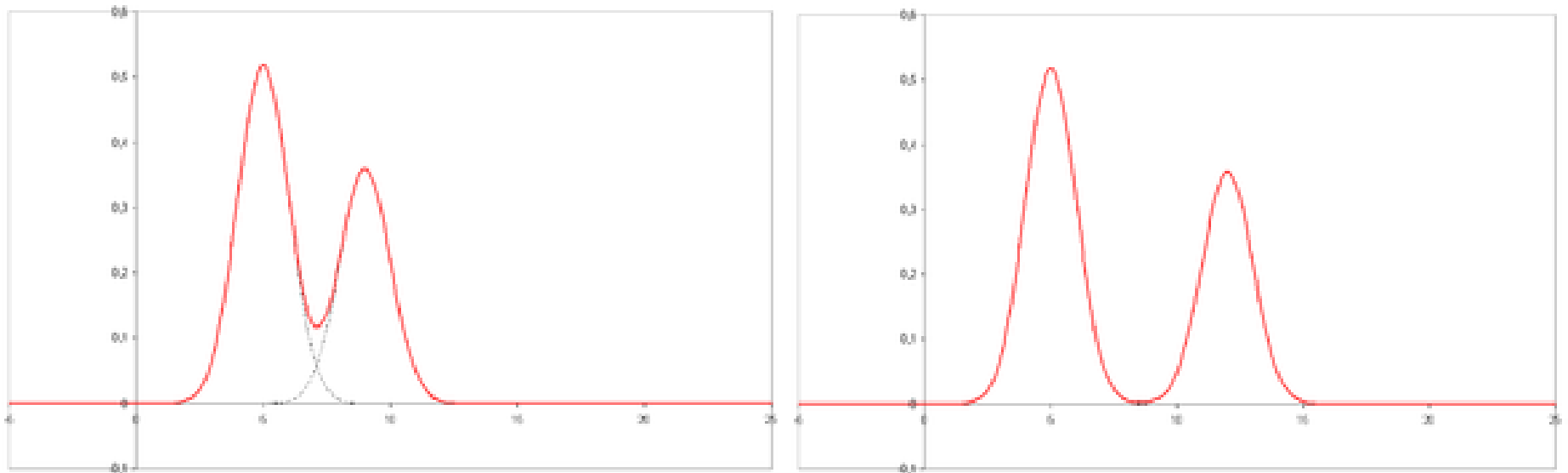
- Gas chromatography is the recent development in the technique for gas analysis capable of analyzing mine gases with accuracy from small samples.

Chromatography

- A collective term for a set of laboratory techniques for the separation of mixtures.
- It is a physical method of separation in which the components to be separated are distributed between two phases,
 - one of which is stationary (stationary phase)
 - while the other (the mobile phase) moves in a definite direction.
- Derived from Greek words *chroma*, color and *graphein* to write.
- It involves passing a mixture dissolved in a "mobile phase" through a *stationary phase*, which separates the analyte to be measured in the mixture based on differential partitioning between the mobile and stationary phases.

Chromatography terms

- A **chromatograph**: an equipment that enables sophisticated separation of gas (gas chromatography) or liquid (liquid chromatography).
- The **analyte**: the substance that is to be separated during chromatography.
- A **bonded phase**: a stationary phase that is covalently bonded to the inside wall of the column tubing.
- A **chromatogram**:
 - The visual output of the chromatograph.
 - Different peaks or patterns on the chromatogram correspond to different components of the separated mixture.



Chromatograms

- Plotted on the x-axis: the retention time and
- Plotted on the y-axis: a signal (in mV) corresponding to the response created by the analytes exiting the system.

Retention time: the characteristic time taken by a particular analyte to pass through the system (from the column inlet to the detector).

- The **mobile phase**:
 - The phase which moves in a definite direction, may be a liquid or gas (GC).
 - It consists of the sample being separated/analyzed and the solvent that moves the sample through the chromatography column (the stationary phase) .
 - The sample during its movement interacts with the stationary phase and gets separated.
- The **effluent**: is the mobile phase leaving the column.
- The **stationary/immobilize phase**: the substance which is fixed in place on the inner wall of the column tubing, viz. **silica layer**.
- The **sample**: is the matter analyzed in chromatography. It may consist of a single component or a mixture of components

Constructional Features

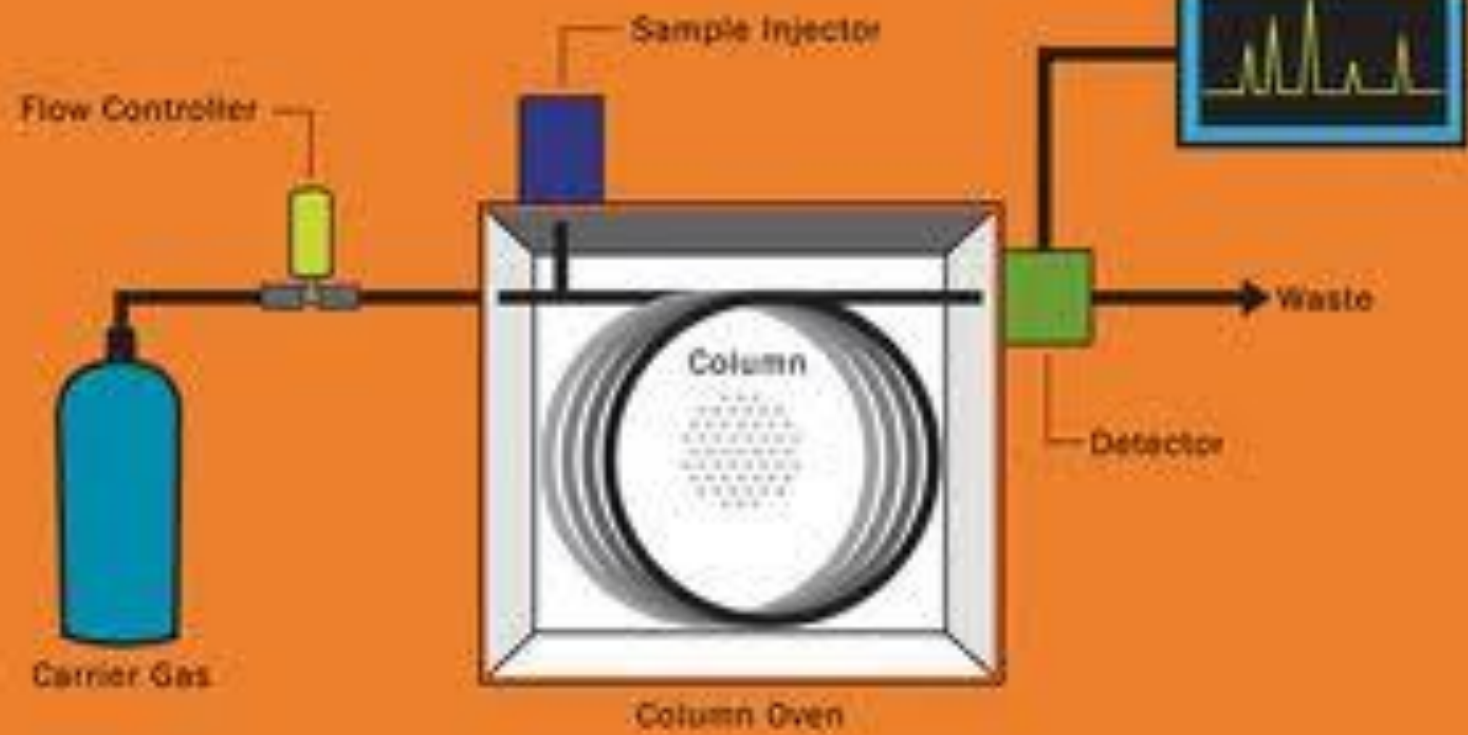
Gas chromatographs essentially comprise

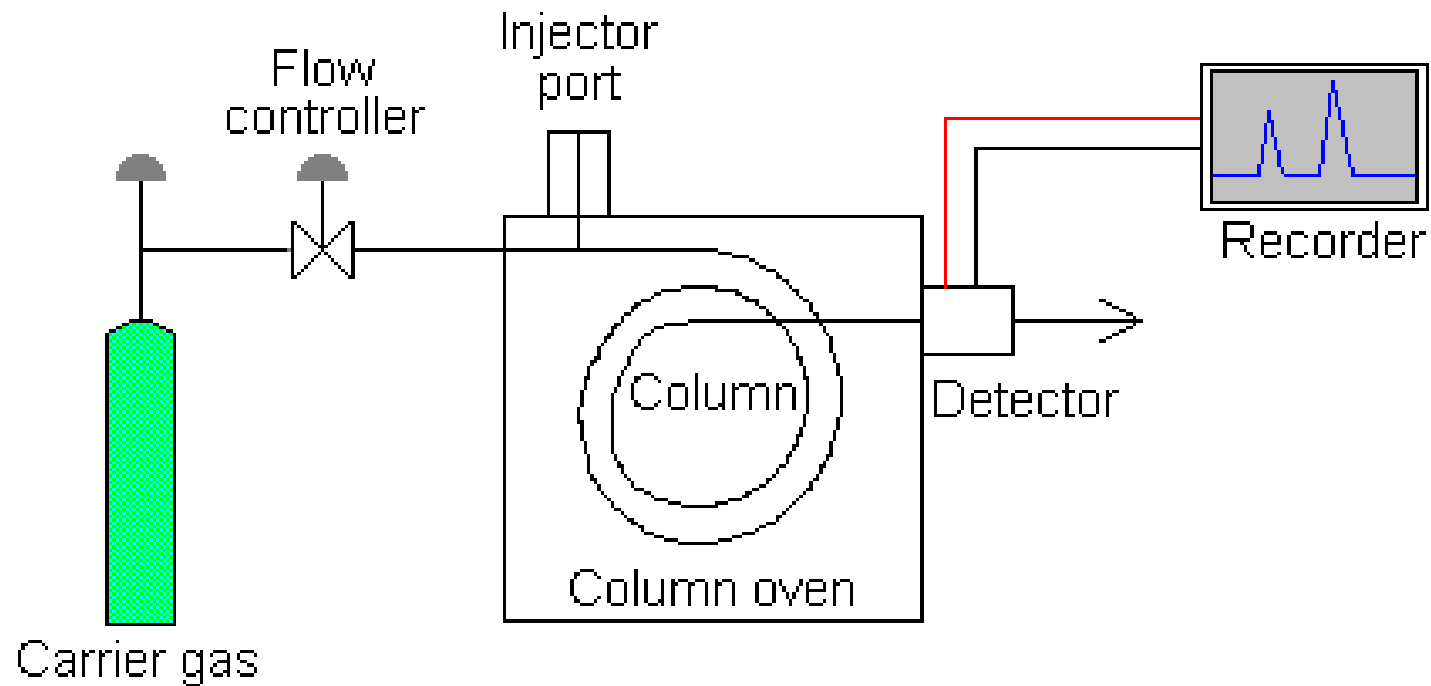
- **A column**, placed in a column oven, containing a stationary phase over which passes a mobile phase of carrier gas from a suitable supply source through a pressure regulator.
- It is the most important part of chromatograph and responsible for separation of individual constituents of a gaseous mixture.
- It contains as adsorbent which retains different gases for different duration in the column.
- For analysis of mine gases like O₂, N₂, CO, NO, CH₄ etc. solid absorbents are commonly used.
- The gaseous component having least retention time comes out first followed by those with longer retention times thus resulting in the separation of individual gases.
- The detector then senses and records the concentration of these individual gases in the form of Gaussian curves (against time), the height of which or preferably the area under which give, by suitable calibration, the conc. of the gas detected.

- **Carrier gas supply**
- **The sample inlet (injectors)** for introducing the air sample to be analyzed into the carrier gas in an injection chamber immediately before the column
- **Detector(s)** provided after the column for recording the concentration of the component gases.
- **A device for data collection, acquisition and processing.**

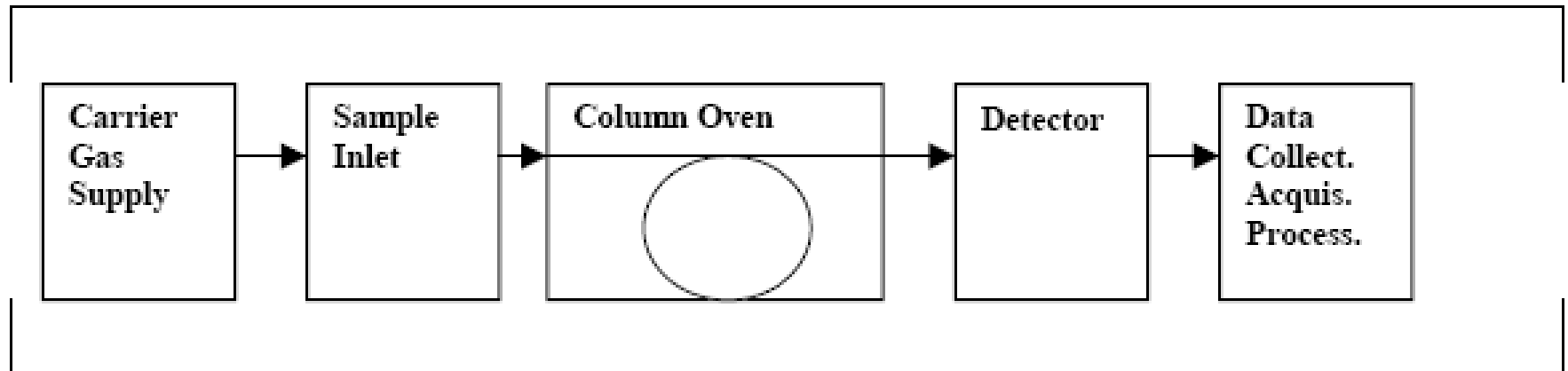








Schematic diagram of a gas chromatograph



Flow diagram of a gas chromatograph

Carrier gas

- An inert gas is used as mobile phase, which is delivered by a gas generator, or a gas cylinder.
- The most common carrier gases are H₂, N₂, He.

Choice of the carrier gas depends on several demands, e.g

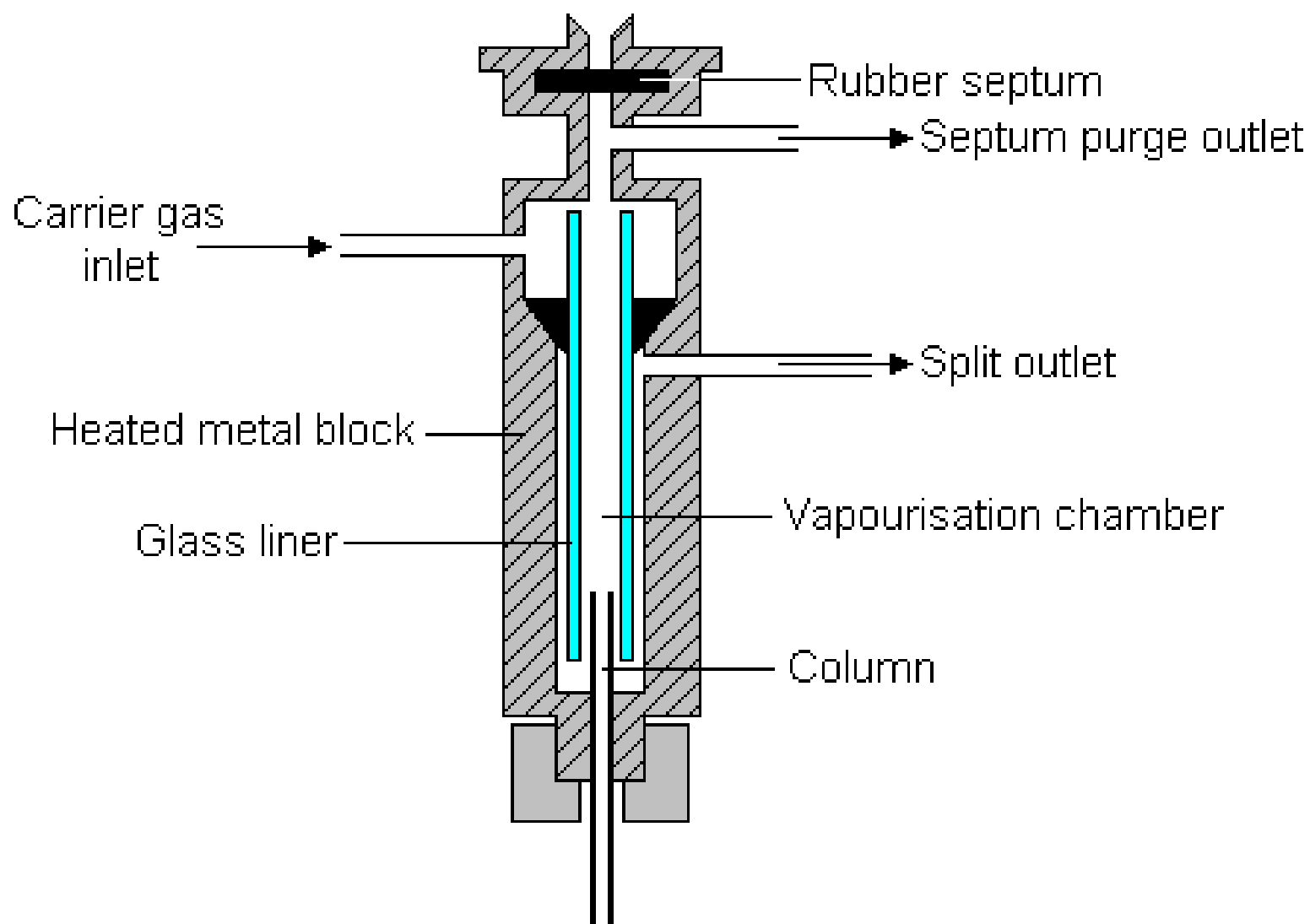
- **Chemically inert**
- **High purity:** must be of high purity and should not contain water & oxygen.
 - traces of water or oxygen may decompose the stationary phase, which leads to column bleeding and finally destruction of the column.
 - special devices for gas purification are often installed prior to the sample inlet.

- **Should have high thermal conductivity** compared to the gas to be detected so that high sensitivity can be obtained with thermal conductivity detectors.
- **Safety reasons:** H₂ is explosive
- **Economic:** N₂ is the cheapest gas
- **Speed and separation efficiency:** Due to its lowest viscosity, H₂ allows to operate the column with the highest mobile phase velocity - and therefore lowest analysis time - at comparable efficiency

Sample injection port

- For optimum column efficiency, the sample should not be too large, and should be introduced onto the column as a "plug" of vapour
- Commonly a microsyringe is used to inject sample through a rubber septum into a flash vapouriser port at the head of the column.
- The temperature of the sample port is usually about 50°C higher than the boiling point of the least volatile component of the sample.
- The injector can be used in one of two modes; split or splitless.

The split / splitless injector

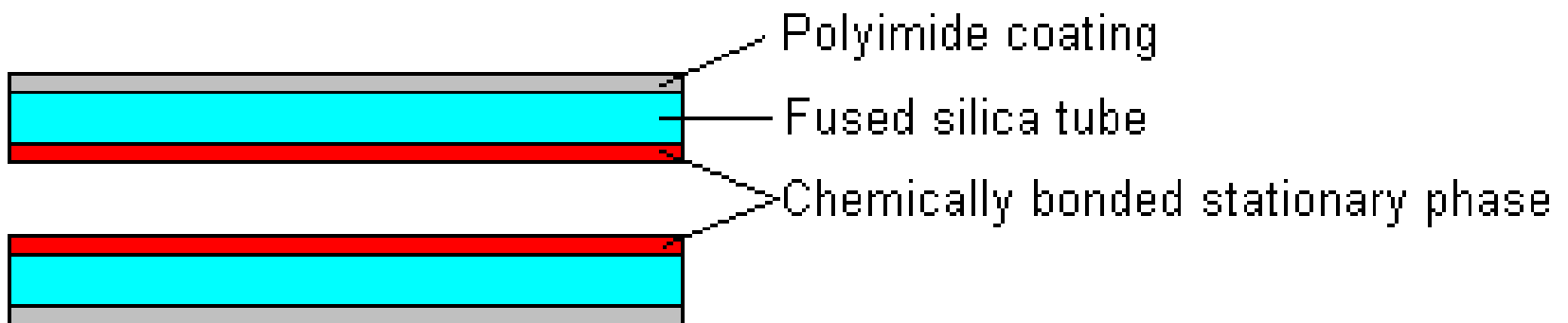


- The injector contains a heated chamber containing a glass liner into which the sample is injected through the septum.
- The carrier gas enters the chamber and can leave by three routes (when the injector is in split mode).
- The sample vapourises to form a mixture of carrier gas, vapourised solvent and vapourised solutes.
- A proportion of this mixture passes onto the column, but most exits through the split outlet.

- There are two general types of column,
 - *packed* and
 - *capillary* (also known as *open tubular*).
- Packed columns contain a finely divided, inert, solid support material (commonly based on *diatomaceous earth*) coated with liquid stationary phase.
- Most packed columns are 1.5 - 10m in length and have an internal diameter of 2 - 4mm.
- Capillary columns have an internal diameter of a few tenths of a millimeter.
- They can be one of two types; ***wall-coated open tubular (WCOT)*** or ***support-coated open tubular (SCOT)***.
- **Wall-coated columns** consist of a capillary tube whose walls are coated with liquid stationary phase. **In support-coated columns**, the inner wall of the capillary is lined with a thin layer of support material such as diatomaceous earth, onto which the stationary phase has been adsorbed.
- SCOT columns are generally less efficient than WCOT columns. However, both types of capillary column are more efficient than packed columns.

- In 1979, a new type of WCOT column was devised - the *Fused Silica Open Tubular* (FSOT) column;
- These have much thinner walls than the glass capillary columns, and are given strength by the polyimide coating.
- These columns are flexible and can be wound into coils.
- They have the advantages of physical strength, flexibility and low reactivity.

Cross section of a Fused Silica Open Tubular Column



Detectors

- There are many detectors which can be used in gas chromatography.
- These detectors comprise thermal conductivity cells or katharometers having elements of heated wire or thermistors.
- As gases of different thermal conductivity pass over the element, it cools to different extent and as a result, the resistance of the element changes.
- The change in resistance is recorded through a bridge circuit and amplifier.

Types of Detectors

- Different detectors will give different types of selectivity.
- A ***non-selective*** detector responds to all compounds except the carrier gas,
- A ***selective detector*** responds to a range of compounds with a common physical or chemical property and
- A ***specific detector*** responds to a single chemical compound.

- For specific gases special types of detectors may be used.
- For examples,
 - electron-capture detectors are very sensitive for NO₂ and halogenated compounds.
 - Flame-ionisation detectors (FID) can be used for hydrocarbons and
 - Microcoulometric detectors for compounds of sulphure.
- It is difficult to analyze a condensable gas like CO₂ or unsaturated hydrocarbons along with other permanent gases present in mine air with a single column.
- In such cases dual columns, one of silica gel for separating CO₂ from air (O₂ and N₂ of air do not get separated here) and the other of molecular sieve for separating the permanent gas components of air such as O₂, N₂, CH₄ etc. are used.